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The Organizational Transmission of AI: The Role of Managers on AI Adoption and Impact

Christos A. Makridis *

April 24, 2026

Abstract

Using longitudinal data from the Gallup Workforce Panel covering more than 30,000 U.S. employees surveyed from 2023 to Q1:2026, I document heterogeneous adoption of AI and its "organizational transmission" within firms. The share of frequent (occasional) AI users grew from under 10% to over 27% (10% to 21%), with post-2024 growth driven by occasional users converting to frequent use rather than new adopters at the extensive margin. Adoption follows the organizational hierarchy—senior leaders reach the low-40s in frequent use by Q1:2026, individual contributors the low-20s—and varies sharply by industry, occupation, and income. Employees reporting that their organization has a clear AI strategy are roughly 27 percentage points more likely to report using AI at least multiple times per week. Strategic clarity is concentrated in organizations that exhibit greater developmental feedback and trust in leadership, suggesting employees infer "strategy" from managerial behavior rather than formal policy. Exploiting within-person variation and a job-switcher design, I show that the relationship between AI use and worker outcomes is strongly contingent on organizational context. Frequent use without a clear strategy carries a near-zero association with engagement and a positive association with burnout; the interaction with clear strategy, however, tends to flip both signs. Perceived clarity also operates through trust in leadership: frequent users in high-trust environments report 0.61–0.72 SD larger perceived productivity gains and tilt usage toward learning rather than routine automation. These patterns are consistent with a model in which managers shape adoption through communication and psychological safety, and benefits are strongest when clarity and trust are jointly high.

Keywords: Generative AI; Technology Adoption; Managers; Trust; Communication; Organizational Complementarities; Panel data. **JEL:** O33; M15; M54.

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1 Introduction

AI has emerged as a transformative technology in the workplace, most recently with a generative dimension (Eloundou et al., 2024). Early experimental evidence indicates that integrating genAI tools can raise worker productivity, improving both the speed and quality of output (Noy and Zhang, 2023; Brynjolfsson et al., 2025). While there are suggestive positive signs of its impact on labor markets (e.g., Johnston and Makridis, 2025), its long-term effects will depend crucially on the speed and breadth of adoption (Brynjolfsson et al., 2021), consistent with historical findings that slow diffusion can delay the realization of productivity gains (Comin and Hobijn, 2010).

Recent evidence shows that generative AI (genAI) uptake has been strikingly fast (Bick et al., 2026). As of late 2024, nearly 40% of U.S. adults have used genAI, with about 23% of employed individuals using it for work at least once in the past week. This diffusion in genAI's first two years has matched or even outpaced that of past major innovations like personal computers and the internet. Despite this rapid overall growth, adoption remains uneven across workplaces and workers. Indeed, many employees are still on the sidelines of the "AI revolution," harboring skepticism about AI's impact on their jobs and its efficacy (Ray, 2024). In addition, roughly a fifth of workers are at least "somewhat concerned" that their jobs may be replaced by AI over the next five years (Makridis, 2026). Such reservations underscore that technological progress does not guarantee universal AI adoption among the workforce—let alone its effective use.

A large literature suggests that managerial influence may be a decisive factor in technology uptake within firms.[1] Managers often serve as technology champions or gatekeepers, shaping their

[1] The industrial psychology literature finds that managers account for roughly 70% of the variance in team engagement (Beck and Harter, 2015).

teams' openness to new tools (Howell and Higgins, 1990; Rogers, 2003; Davenport, 2018). In addition, management practices—embodied through individual managers—are integral to unlocking the benefits of technology (Bresnahan et al., 2002; Bloom et al., 2012; Aral et al., 2012). This influence likely extends to shaping employees' willingness to experiment with innovations. Only about one-third of U.S. workers report that their organization has started integrating AI (Den Houter, 2024)—although another study places this as low as 18% (McElheran et al., 2024)—and a mere 15% strongly agree that their employer has communicated a clear AI strategy in 2024, up now to 26%. When such a plan is in place, employees are nearly five times more likely to feel comfortable using AI. These patterns suggest that clear managerial communication and vision are critical in converting top-down AI investment into broad-based adoption on the ground. Trust in management is another potentially crucial element: employees who lack trust in their leaders may be especially reluctant to embrace new technologies and integrate them into their workflow. In short, leadership may make or break the "last-mile" of technological diffusion within organizations.

The primary contribution of this paper is to study how managers shape the internal diffusion of AI and how that diffusion maps into employee outcomes. Using longitudinal data from the Gallup Workforce Panel covering more than 30,000 U.S. employees surveyed from 2023 to Q1:2026, I document rapid growth in workplace AI use alongside systematic heterogeneity that points to an organizational transmission mechanism. In the aggregate, frequent use rose from under 10% in 2023 to over 27% by Q1:2026, and occasional use expanded from roughly 10% to 21%. Within-person flow decompositions indicate that post-2024 growth is driven primarily by existing occasional users converting to frequent use—the Occasional → Frequent transition rate reaches one in three by mid-2025—rather than by new adopters entering at the extensive margin.

But more importantly, AI diffusion is driven substantially by structured organizational conditions and managerial channels rather than only by individual characteristics. Adoption follows the hierarchy: senior leaders reach the low-40s in frequent use by Q4:2025, with supervisors and project managers close behind, while individual contributors remain in the low-20s. Even the public sector has seen a rise: federal workers, despite the most gradual trajectory, close much of the gap with the private sector by Q1:2026. AI use also remains widespread across the income distribution, though a widening gradient on the frequent-use margin (the \$180K+ group reaches nearly 46% by Q1:2026 versus roughly 19% for workers earning under \$48K) raises concerns about unequal access to productivity gains. Cross-sectionally, the industry and occupation gradients align with task-level exposure: Technology/Math leads at roughly 60% frequent use, followed by FIRE and Consulting in the mid-40s, while Hospitality/Travel/Food lags at 15%. Growth over late 2025 is concentrated in intensification—frequent use rising, occasional use flat or declining—consistent with a reallocation from experimentation toward embedded adoption.

Motivated by the large evidence linking management practices with productivity and technology adoption, I study the cross-sectional determinants of AI adoption within organizations. I find that perceived clarity of an organization's AI strategy is the strongest and most economically meaningful correlate of frequent adoption. In a saturated specification conditioning on demographics, individuals who report perception of a clear AI strategy exhibit a 27 percentage point higher probability of frequent use (0.268, s.e. 0.012), while most individual workplace-practice coefficients attenuate to zero. Trust in leadership is the one exception, retaining independent predictive content (0.016, s.e. 0.007). Examining what predicts strategic clarity, feedback is the most robust correlate (0.040, s.e. 0.006 in the saturated model), with connectedness and trust retaining positive but attenuated coefficients—suggesting that employees infer "AI strategy" from the quality of day-to-day managerial communication rather than from formal policy announcements.

Exploiting within-person variation over time, occupation × year controls, and a job-switcher design, I show that the association between AI use and outcomes is strongly contingent on organizational context. In pooled specifications, frequent AI use without a clear strategy carries a near-zero or negative association with engagement (−0.014 SD) and a positive association with burnout (0.115 SD); the interaction with clear strategy is 0.213 SD for engagement and −0.074 SD for burnout, implying that the sign of the relationship flips with organizational context. With person and wave fixed effects, the engagement complementarity attenuates but persists (0.064 SD for frequent use, 0.051 SD for occasional use), and strengthens further with SOC3 × time controls (0.071 and 0.083 SD, respectively). Among job switchers—where variation in organizational environment is mechanically linked to employer transitions—the burnout and satisfaction complementarities are sharpest: frequent use without clear strategy raises burnout by 0.272 SD, but the interaction offsets most of this penalty (−0.234 SD). Across the Q12 components, complementarities concentrate in access to materials, developmental encouragement, and social connection. The engagement complementarity for frequent use is strongest among non-White workers (0.134, s.e. 0.053), men (0.079, s.e. 0.038), and college-educated respondents (0.061, s.e. 0.026). Trust in leadership further amplifies the returns: within-person estimates show that frequent users in high-trust environments report 0.61–0.72 SD larger perceived productivity gains, and high-trust adopters tilt usage toward learning and idea generation rather than routine automation. These results are consistent with a stylized theoretical model in which workers adopt and experiment with AI when expected gains exceed perceived costs, and managers influence both margins. Clear communication reduces uncertainty about permissible use and raises the expected value of learning-by-doing; trust and psychological safety reduce the perceived downside risk of experimentation. The model predicts complementarities—adoption and its benefits should be strongest when clarity and psychological safety are jointly high—precisely as observed in the panel evidence.

This paper contributes primarily to the empirical management and organizational economics literature by providing new evidence on how managerial practices and workplace culture shape the within-firm diffusion of AI and its consequences for workers. A large body of research establishes that management quality is a central source of productivity differences across firms and countries (Bloom and Van Reenen, 2010; Bloom et al., 2013, 2017), and that the returns to digital technologies are larger when firms pair information technology (IT) with complementary organizational practices and skills (Bresnahan et al., 2002; Bloom et al., 2012; Aral et al., 2012). AI has become the next frontier of the digital revolution, spurred by the release of ChatGPT in late 2022. Despite rapid growth in generative AI use (Bick et al., 2026) and accumulating evidence that it can raise output in specific work settings (Noy and Zhang, 2023; Brynjolfsson et al., 2025; Dell'Acqua et al., 2026, 2025; Cui et al., 2025), there is limited direct evidence on which organizational mechanisms govern whether employees experiment with AI, how quickly usage diffuses across colleagues, and whether adoption translates into better (or worse) workforce outcomes. Motivated by the "automation versus new tasks/augmentation" perspective (Autor, 2015; Acemoglu and Restrepo, 2019; Autor et al., 2022), my central premise is that the realized impact of AI depends not only on tool availability, but also on the managerial infrastructure that reallocates tasks, legitimizes experimentation, and mitigates perceived downside risks.

A key contribution of the paper is measurement and identification at the organizational layer. The substantial within-firm heterogeneity in adoption and in the correlates of adoption matters for two reasons. First, it provides micro-level evidence consistent with the view that general-purpose technologies require intangible complements that are costly to build and slow to diffuse (Brynjolfsson et al., 2021). Second, it suggests a mechanism through which AI could widen performance gaps across firms: organizations that pair AI with strong managerial practices and high-trust climates appear better positioned to translate experimentation into sustained adoption and improved worker outcomes, while poorly aligned firms risk low uptake and muted (or adverse) effects. This mechanism complements firm-level evidence on who adopts AI and where (McElheran et al., 2024), by opening the black box of internal diffusion and showing that managerial conditions help explain why similar technologies yield very different realized impacts across workplaces.

The paper also contributes to the technology adoption literature by shifting emphasis from individual-level acceptance to organizational design and leadership. Classic acceptance frameworks center perceived usefulness and ease of use (Davis, 1989; Venkatesh et al., 2003). However, a growing body of evidence suggests that the returns to new digital technologies depend critically on organizational complements—especially the availability of complementary skills—rather than on user perceptions alone (Tambe, 2014). In the context of digital transformation, employees who trust their managers may be more willing to follow guidance to try unfamiliar technologies and less fearful of negative consequences (e.g., job loss) from automation. For example, Makridis and Han (2021) finds that the positive effects of technological change on employee empowerment and satisfaction are strongest in workplaces characterized by high trust in management and more directive (structured) managers. In other words, a trusting and well-led environment can help workers embrace and benefit from new technologies, rather than resist them. My results instead show that managerial communication and workplace norms—whether leaders provide clear guidance, whether employees trust managers, and whether feedback is candid and developmental—are first-order determinants of whether workers view AI experimentation as safe and worthwhile. This emphasis on internal learning and assimilation connects the adoption literature to organizational change and trust research (Morgan and Zeffane, 2003) and to psychological safety and learning in organizations (Edmondson, 2019), and it is consistent with evidence that implementation outcomes hinge on work practices that facilitate employee learning during technology roll-outs (Avgar et al., 2018). These findings suggest that "soft" cultural factors materially condition the speed and breadth of diffusion of a new tool inside firms and explain the relatively high failure rate.[2]

A further contribution is to the literature on complementarities between technology and organizational practices: investments deliver larger productivity gains when firms simultaneously adopt complementary workplace organization and incentive systems (Bresnahan et al., 2002; Aral et al., 2012). I identify specific complements—strategic clarity about AI use, trust/psychological safety, and feedback-rich management—that predict both adoption intensity and the direction of workforce outcomes (e.g., engagement, satisfaction, burnout). This evidence aligns with the "productivity J-curve" logic: realizing gains from a general-purpose technology often requires complementary investments in processes and organizational capital that may precede observable productivity improvements (Brynjolfsson et al., 2021). Similarly, I quantify how cultural/managerial complements can operate as force multipliers for AI diffusion and impact.

Finally, the paper offers a theoretical contribution by developing a stylized model of the "organizational transmission" of AI that disciplines the empirical analysis. Building on organizational economics models of authority, communication, and coordination (Garicano, 2000; Dessein, 2002; Alonso et al., 2008; Crémer et al., 2007) and on relational incentives and credibility (Baker et al.,

[2]A widely cited "MIT study" is often summarized as finding that "95% of AI pilots fail." In fact, the underlying source is a Project NANDA report that documents a sharp pilot-to-production drop-off in a *selected* sample of enterprise genAI initiatives (Challapally et al., 2025). The report's "95%" figure refers to organizations in its dataset reporting *no measurable return / no measurable P&L impact* from genAI efforts (and it separately labels this pattern a "95% failure rate"), not to a population-wide estimate of technical failure. The evidence is drawn from a systematic review of 300+ publicly disclosed initiatives, interviews with representatives from 52 organizations, and surveys of 153 senior leaders at industry conferences; the authors caution that deployment and "success" rates are directionally informative, based on interviews rather than audited reporting, and that definitions vary across organizations. Even with these limitations, the reported attrition from pilots to workflow-integrated production systems indicates a nontrivial implementation failure rate, consistent with the view that organizational complements (integration, change management, and employee adoption) are central determinants of whether pilots scale.

2002; Gibbons and Henderson, 2012), I build a simple model that formalizes how managers shape adoption through two interdependent choices: (i) complementary investments that lower the friction of integrating AI into workflows and (ii) the clarity and credibility of communication that governs whether employees perceive experimentation as valuable and safe. The model yields testable complementarities—adoption and its benefits should be strongest where clarity and psychological safety are jointly high—and provides a mechanism linking workplace culture to the equilibrium diffusion of a new technology within firms over time and space.

2 Modeling the Organizational Transmission of AI

This section provides a stylized framework that (i) clarifies the mechanisms through which management shapes AI adoption and (ii) delivers testable implications used to organize the empirical analysis. The model builds on core ideas in organizational economics that treat within-firm communication as a design choice that affects coordination and the use of dispersed information (e.g., Dessein, 2002; Alonso et al., 2008; Crémer et al., 2007), and on the view that many managerial "promises" are sustained by informal (relational) contracting rather than verifiable formal contracts (Baker et al., 2002; Gibbons and Henderson, 2012).

2.1 Setup

A firm consists of a unit mass of ex ante identical workers and a manager. Workers choose an intensity of AI usage $a \in [0, 1]$, interpreted as the fraction of tasks completed with AI assistance (Eloundou et al., 2024). Using AI increases effective performance but requires experimentation and adjustment, generating private costs. Let $E \in [0, 1]$ denote the firm's "embedding" investment (training, workflow redesign, IT integration, and complementary inputs). Let S denote the clarity/precision of managerial communication about AI use (the "AI strategy").

The per-task gross return to AI is

$$B(E, S), \tag{1}$$

with $B_E > 0$, $B_S \geq 0$, and $B_{ES} > 0$ for complementarity between embedding and communication. Embedding investments are more productive when the organization has a shared, interpretable language and guidance about use-cases, standards, and responsibilities (Crémer et al., 2007).

Let θ index psychological safety/trust in the work environment. Higher θ reduces the perceived downside risk of experimentation and lowers the marginal cost of adoption, i.e. $c_{a\theta} < 0$. Worker costs are given by

$$c(a; \theta), \tag{2}$$

with $c_a > 0$, $c_{aa} > 0$, $c_{a\theta} < 0$, and $c(0; \theta) = 0$; θ captures the extent to which employees view experimentation (e.g., early mistakes) as safe and not career-damaging.

At the implementation stage, the firm chooses embedding E, and the manager chooses communication clarity S and takes actions shaping psychological safety θ. Workers then choose a after observing (E, S, θ). Conceptually, S corresponds to the informativeness/precision of internal communication and guidance: clearer communication improves alignment and coordination but can be constrained by organizational design and incentive conflicts in communication (Dessein, 2002; Alonso et al., 2008). The parameter θ captures the credibility of informal assurances around experimentation and learning, which are typically noncontractible (Baker et al., 2002).

Given (E, S, θ), a worker chooses a to maximize

$$\max_{a \in [0,1]} \; a \cdot B(E, S) - c(a; \theta). \tag{3}$$

An interior optimum satisfies

$$B(E, S) = c_a(a^*; \theta). \tag{4}$$

Lemma 1 (Adoption comparative statics)**.** *Suppose $c(\cdot; \theta)$ is strictly convex and $B(E, S)$ satisfies $B_E > 0$, $B_S \geq 0$, $B_{ES} > 0$. Then the equilibrium adoption intensity $a^*(E, S, \theta)$ is weakly increasing in E, S, and θ. On interior regions where $a^*(E, S, \theta) \in (0, 1)$, adoption is strictly increasing in E and θ, weakly increasing in S, and that the marginal effect of embedding is larger when strategy clarity is higher (i.e., $\partial^2 a^*/\partial E \partial S > 0$ under standard regularity conditions).*

Lemma 1 formalizes the idea that adoption is not "plug-and-play": both a higher return environment (via E and S) and lower experimentation costs (via θ) raise equilibrium AI usage. A closed-form example is provided in Section A.1.1 of the Online Appendix.

The central empirical implication concerns complementarities: the marginal effect of AI usage on performance and well-being depends on the managerial environment. Let W denote an organizational outcome of interest (e.g., productivity, engagement, burnout). We remain agnostic about the full mapping from (a, E, S, θ) to outcomes, but impose a complementarity structure:

$$\frac{\partial^2 W}{\partial a \partial \theta} > 0, \qquad \frac{\partial^2 W}{\partial a \partial S} > 0. \tag{5}$$

Intuitively, psychological safety increases the returns to experimentation with AI, while clear communication improves coordination and reduces role ambiguity, increasing the payoff to AI usage. This aligns with a broader literature on complementarities between technology adoption and organizational practices (Milgrom and Roberts, 1990; Bresnahan et al., 2002).

Proposition 1 (Environment–adoption complementarity). *Under (5), the marginal effect of AI use on outcomes is larger in high-θ and high-S environments. In particular, holding E fixed, the slope of W with respect to a is increasing in θ and S.*

Section A.1.1 of the Online Appendix provides proofs and discusses how alternative microfoundations (e.g., communication as signal precision or "language" constraints) map into S, following Crémer et al. (2007) and strategic communication in Dessein (2002); Alonso et al. (2008).

The model delivers a set of hypotheses that guide the empirical analysis. Throughout, ClearAI proxies for S, trust/psychological safety proxies for θ, and embedding E is treated as partially observed (and absorbed by fixed effects and controls where appropriate).

H1 (Adoption levels). AI adoption intensity is weakly increasing in embedding E, communication clarity S, and psychological safety θ (Lemma 1).

H2 (Complementarity in outcomes). The marginal association between AI usage and worker outcomes is more positive (or less negative) when ClearAI is high and when trust/psychological safety is high (Proposition 1).

H3 (Task interdependence amplifies the value of clarity). The outcome complementarity in H2 is stronger in jobs with higher task interdependence/coordination demands. The interaction between AI use and ClearAI is increasing in an occupation-level interdependence measure.

H4 (Credibility: trust strengthens the effect of clarity). The AI use × ClearAI interaction is larger when trust/psychological safety is high (a triple complementarity), consistent with interpreting θ as capturing the credibility of informal assurances around experimentation.

H5 (Composition of use). Conditional on adopting AI, ClearAI shifts usage toward more standardized, workflow-embedded applications (where alignment is valuable), while trust shifts usage toward more exploratory applications where experimentation costs are salient.

H6 (Role clarity crossover). AI usage decreases role clarity when ClearAI is low (unmanaged adoption increases ambiguity), but increases role clarity when ClearAI is high (guidance clarifies expectations). Empirically, this predicts a sign change across clarity regimes.

3 Data and Measurement

3.1 Gallup Workforce Panel

I primarily leverage data from the Gallup Workforce Panel, a longitudinal and nationally-representative survey of U.S. adults. The panel is probability based and surveys workers (18+) about workplace experiences, management, and well-being, with core questions repeated and new items added on emerging topics such as AI. For instance, the Q2 2024 wave was a web survey of 19,043 working adults drawn from this panel, with sampling weights calibrated to national demographics.

I restrict the sample to employed respondents between ages 20 and 65. Although each pre-2025 wave includes roughly 20,000 respondents per quarter, AI questions were only asked in the second quarter; starting in Q2:2025, AI questions are asked quarterly. In each quarter, however, there are roughly 10,000 respondents to AI questions. This gives 33,743 unique workers and, depending on the specification, at least 78,939 person-wave observations across 2023 to Q1:2026.[3]

3.1.1 AI Measurement

The primary outcome is an indicator of AI adoption at the individual level. In particular, I use answers to the question How often do you use artificial intelligence in your role," which include: (a) daily, (b) a few times a week, (c) a few times a month, (d) few times a year, (e) once a year, (f) less often than once a year, and (g) never. I consolidate these answers into three groups: often (daily and a few times a week), sometimes (a few times a month, a few times a year), and never (once a year, less often than once a year, and never). Before asking this question, respondents are given a definition of AI: Artificial intelligence is when a computer does or enhances a task that is normally performed by a person." Figure 1 visualizes the time path of adoption and shows two distinct phases. From 2023 to 2024, adoption rises gradually and asymmetrically: frequent use edges up from 9.7% to 12.1%, while occasional use expands more sharply from 9.7% to 16.0%, suggesting that early diffusion was driven primarily by workers experimenting with AI rather than integrating it into regular workflows. The trends intensify markedly in 2025 and into early 2026. Frequent use climbs from 19.0% in Q2:2025 to 22.3% in Q3, 25.0% in Q4, and reaches 27.4% by Q1:2026. Occasional use, by contrast, rises to around 20% by Q2:2025 and remains comparatively flat thereafter, hovering between 18.8% and 21.3% through Q1:2026. Theere is a divergence between frequent and occasional use after Q2:2025, suggesting that the extensive

[3]Figure A.1 summarizes the longitudinal variation. The mean number of waves per worker is 2.41, and over 60% of respondents are observed in at least two waves. The exact number of respondents in the sample varies based on whether certain variables, such as perceptions of the workplace or employee engagement responses, are available for a respondent, and thus oscillates.

margin of experimentation had largely saturated by mid-2025, with subsequent growth driven by workers deepening and routinizing their use rather than new adopters entering at the occasional margin.[4]

These patterns are broadly consistent with, but extend, the evidence in Bick et al. (2026). They report that as of late 2024, roughly 23% of employed adults used AI for work at least once in the prior week and about 9% used it daily. In this data, the 2024 frequent share (daily or a few times per week) is 12.1% and the occasional share is 16.0%, implying about 28% of workers use AI at least occasionally—broadly in line with their estimate once definitional differences are accounted for.[5] Besides the sample size and breadth of questions, a key advantage of the present data is the ability to track diffusion for the same individual over time. Figure A.2 in the Online Appendix decomposes the aggregate rise in frequent use into within-person flows. The Occasional → Frequent transition rate is the dominant engine: roughly one in five occasional users converted to frequent use by 2024, rising to one in three by Q2:2025 before moderating to around one in four by Q1:2026. By contrast, the Never → Occasional and Never → Frequent rates are comparatively stable, suggesting the extensive margin of new adopters had largely plateaued by mid-2025.[6]

By contrast, the adoption levels in Hartley et al. (2025) are less directly comparable for potentially several reasons. First, the core adoption concept is broader (e.g., "ever used genAI at work since tools became available"), which mechanically yields higher rates and is more sensitive to sporadic experimentation and sample composition. Second, the distinction between nationally

[4]Table A.2 in the Online Appendix provides descriptive statistics on the sample by year.

[5]Their headline "weekly" measure sits between the Gallup "frequent" margin (daily or a few times per week) and the combined frequent-plus-occasional measure, which likely explains the modest level differences.

[6]That conditional rates moderate after Q2:2025 yet aggregate frequent use continues to rise reflects the expanding pool of occasional users—a stable transition rate applied to a larger base still generates more absolute upgraders each wave. The within-person flows thus point to deepening routinization among existing adopters, rather than an influx of new ones, as the primary driver of post-2024 growth.

representative designs and low-barrier online sampling is becoming first-order for measurement: Westwood (2025) shows that LLM-driven "synthetic respondents" can evade standard attention and quality checks at extremely high rates and can be deployed strategically to bias survey estimates. This evidence reinforces the value of probability-based sampling frames for tracking AI adoption, especially as LLM capabilities grow and the incentives to contaminate online panels rise.

Table A.1 shows that 42.9% change their AI use category at least once; as a share of all unique workers (including single-wave respondents), 26.8% ever switch. Considering adjacent-wave transitions, there are 46,067 valid t-1 $\rightarrow$ t comparisons. Of these, 13.5% are adoptions (Never $\rightarrow$ Occasional/Frequent) and 5.5% are cessations (Occasional/Frequent $\rightarrow$ Never), yielding an adoption-to-cessation ratio of 2.46. Row percentages from the transition matrix indicate persistence and directional movement. Among those previously Never, 75.4% remain Never, 16.2% move to Occasional, and 8.4% move to Frequent. Among those previously Occasional, 54.6% remain Occasional, 27.9% move to Frequent, and 17.5% revert to Never. Among those previously Frequent, 77.3% remain Frequent, 15.5% move to Occasional, and 7.1% revert to Never. Diffusion outpaces "abandonment" over the sample and movements up the intensity ladder are common.

In some specifications, I also make use of another question on AI concern and expectations, namely: "How likely is it that the job you have now will be eliminated within the next five years as a result of new technology, automation, robots or artificial intelligence?" Respondents can choose among "very likely," "somewhat likely," "not too likely," "not at all likely," and "don't know or doesn't apply." I classify the last group as missing, defining an indicator for fear that AI will replace the respondent's job based on reporting "very likely" or "somewhat likely".

3.2 Workplace Practices

I measure workplace (management) practices based on z-scores of responses to the degree of agreement with the following on a 1-5 scale: (a) I feel connected to my organization's culture, (b) My organization consistently delivers on the promises it makes to customers, (c) I trust the leadership of this organization, (d) My organization cares about my overall wellbeing, (e) I have received meaningful feedback in the past week, and (f) I am treated with respect at work.

3.3 Employee Outcomes

The primary employee outcome is the concept of employee engagement, which draws on the "Q12" indicators—a set of 12 questions validated by Gallup as defining employee engagement: (1) I know what is expected of me at work, (2) I have the materials and equipment I need to do my work right, (3) At work, I have the opportunity to do what I do best every day, (4) In the last seven days, I have received recognition or praise for doing good work, (5) My supervisor, or someone at work, seems to care about me as a person, (6) There is someone at work who encourages my development, (7) At work, my opinions seem to count, (8) The mission or purpose of my organization makes me feel my job is important, (9) My associates or fellow employees are committed to doing quality work, (10) I have a best friend at work, (11) In the last six months, someone at work has talked to me about my progress, (12) This last year, I have had opportunities at work to learn and grow. To generate a reliable average, I restrict averages to respondents answering at least 10 or the 12 questions, but since that generates over 20% of missing values, I focus the baseline measure of employee engagement on four of the Q12 items with the broadest coverage: (Q1) expectations, (Q3) do what I do best, (Q6) encourages development, and (Q7) opinions count. I also focus on two additional outcome metrics: burnout and job satisfaction. Burnout is measured as follows: "Please indicate how often each of the following statements is true of your primary job (i.e., the job where you spend the most time working). I feel burned out at work: (a) Always, (b) Very often, (c) Sometimes, (d) Rarely, (e) Never." Job satisfaction is measured (ranked on a 1 to 5 scale) as follows: "How satisfied are you with your place of employment as a place to work?"

3.4 Individual Controls

The primary controls include: age, male, race (White, Black), full time and part time status, having children, and married. I also observe the employee's annual income bin, education, industry, and occupational classification (using a mapping of job titles to SOC-codes), and tenure, although these latter variables are used only in robustness checks due to comparatively more missing values.

4 Descriptive Evidence on AI Utilization

4.1 Public v. Private Sector

Figure 2 compares AI adoption across federal, state/local, private, and non-profit workers. In 2023, frequent use is low and fairly similar across sectors, ranging from around 7% among non-profits to 11% among federal and private workers; occasional use is lowest among federal workers and highest among non-profits. By 2024, adoption rises on both margins, with the increase in occasional use particularly pronounced among state/local and non-profit workers, both of whom reach the high teens, while the occasional share among private sector workers lags behind. The sectors begin to diverge more sharply from Q2:2025 onward. Private and non-profit workers pull ahead on frequent use, both climbing to around 21%, while federal and state/local workers remain closer to the mid-teens. State/local workers stand out for sustaining the highest occasional use share throughout, consistently in the mid-to-high 20s, suggesting a large pool of experimenters who have not yet converted to routine use. By Q1:2026, the private sector leads on frequent use at roughly 29%, with non-profits close behind; federal workers, despite the most gradual trajectory, close much of the gap by Q1:2026, reaching the mid-20s alongside state/local workers.

These patterns suggest that non-profits and government are not persistent laggards in genAI adoption, which is striking given broader concerns about slow technology diffusion outside the private sector. A plausible reason is that genAI is relatively easy to trial compared to traditional machine learning, and that coordination-intensive tasks common in public and mission-driven work provide natural use cases. The remaining gap is in *intensification*: the private sector converts experimentation into frequent use more quickly, consistent with fewer compliance constraints and stronger organizational scaffolding for embedding new tools into workflows.

4.2 Work Type

Figure 3 examines adoption by organizational role and shows a hierarchy gradient that widens over time. In 2023, frequent use is modest across the board—about 16% among senior leaders, 13% among supervisors, 11% among project managers, and 8% among individual contributors—while occasional use is roughly 17%, 15%, 14%, and 7%, respectively. By 2024, adoption rises in every role: frequent use reaches about 20% for senior leaders and roughly 12–13% for supervisors and project managers, with individual contributors around 11%. Occasional use also increases, with especially pronounced growth among senior leaders (high-20s) and project managers (low-20s).

These patterns accelerate markedly from 2025 onward, particularly at the top of the hierarchy. In Q2:2025, frequent use climbs to roughly 32% for senior leaders and about 24–25% for supervisors, with project managers in the low-20s and individual contributors in the mid-teens; occasional use is similarly elevated, near 29% for senior leaders and the mid-20s for supervisors and project managers. In Q3:2025, frequent use rises further for senior leaders (mid-to-high 30s) and project managers (high-20s), while individual contributors remain below 20%. By Q4:2025, the gradient is most pronounced: senior leaders reach the low-40s in frequent use, project managers the mid-30s, supervisors around 30%, and individual contributors the low-20s, while occasional use remains elevated but does not increase proportionally. By Q1:2026, senior leaders extend their lead further, while the gap between supervisors and project managers on one hand and individual contributors on the other remains wide. Diffusion is thus driven disproportionately by shifts toward frequent, embedded use, and that intensification is strongest in leadership and coordination-intensive roles, consistent with models that feature coordination costs (Lee and Makridis, 2023).

4.3 Household Income

Figure 4 tracks AI adoption by household income and reveals both rapid overall growth and a widening income gradient, particularly on the frequent-use margin. In 2023, adoption is low and compressed across brackets: frequent use clusters between roughly 8-11% for most groups, while occasional use ranges from around 8% at the bottom to approximately 15% for the highest earners, leaving only a modest gap. By 2024, frequent use edges into the low-to-mid teens for most groups, with the $180K+ group beginning to pull ahead, and occasional use rises similarly across the distribution with differences remaining relatively contained.

The divergence sharpens considerably from Q2:2025 onward. On the frequent-use margin, the \$180K+ group accelerates steeply, reaching roughly 34% by Q2:2025 and nearly 46% by Q1:2026. By Q1:2026, the \$120K-\$179.9K group reaches around 33%, \$90K-\$119.9K sits near 26%, and \$48K-\$89.9K hovers around 20%. The bottom bracket—households earning less than \$48K—reaches only roughly 19 frequent use by Q1:2026, meaning the gap between the top and bottom of the distribution has grown substantially since 2023. The occasional-use panel tells a different story: the income gradient there is considerably flatter across the entire period. This divergence between the two panels suggests that higher earners are not simply using AI more casually—they are converting to heavy, regular use at a substantially faster rate than lower-income households.

4.4 Industries and Occupations

Starting with industries, Panel A in Figure 5 documents substantial cross-sectional heterogeneity in the intensity of workplace AI use as of Q1:2026. Technology/Math is the clear outlier on frequent use: roughly 60% report using AI frequently, compared to about 22% who use it occasionally. A second tier of information-intensive industries—notably FIRE and Consulting—also exhibits high adoption, with frequent-use rates around 43–47% and occasional-use rates around 26–27%. Education sits just below, with frequent use near 30% and occasional use similarly elevated, consistent with broad integration across both margins. Arts/Entertainment/Media and Health Care form a middle tier, with frequent-use rates in the high-20s, though Health Care is notable for a relatively compressed occasional-use share of around 17%, suggesting that use is concentrated among habitual rather than experimental adopters. By contrast, several customer-facing or operational sectors sit lower in the distribution. Retail/Sales and Industrial/Extraction have frequent-use rates around 20–21%, while Hospitality/Travel/Food is the laggard, with frequent use near 15% and occasional use similarly modest. The cross-section is consistent with what we know about the industries that have extensive AI exposure: industries centered on information production and knowledge work show deeper AI integration, while sectors with more in-person service and manual task content exhibit more limited adoption (Eloundou et al., 2024).

Panel B complements the cross-section by highlighting how quickly intensity is changing within industries, comparing the average of Q2:2025 and Q3:2025 to the average of Q4:2025 and Q1:2026. Growth is concentrated in frequent use: FIRE and Technology/Math lead with gains of roughly 9–10 percentage points, followed by Health Care and Arts/Entertainment/Media at around 7–8 pp and Consulting and Education at roughly 5–7 pp. Even industries with moderate baseline levels in Panel A therefore display rapid deepening of use. Changes in occasional use are more mixed and, in several cases, negative—Technology/Math shows the largest decline in occasional use (roughly −5 pp) alongside the largest gain in frequent use, consistent with a reallocation from occasional experimentation toward routinized adoption. A similar pattern, though smaller in magnitude, appears in Industrial/Extraction and Other, where occasional use falls modestly while frequent use rises. By contrast, Retail/Sales stands out for broad-based expansion, with gains on both margins. Hospitality/Travel/Food remains the slowest-moving sector, with near-zero growth on either margin, reinforcing its position at the bottom of the distribution.

Turning to occupations, Figure 6 reports both the cross-sectional distribution of AI-use intensity (Panel A) and the change in those shares over 2025 (Panel B). Panel A shows a pronounced gradient by task content as of Q1:2026. Computer/Math occupations are the clear leader on routinized use, with roughly 57% reporting frequent AI use and about 24% occasional use. Business/Finance is also high, with frequent use near 44% and occasional use around 30%. A broad middle tier—including Life/Physical/Social Science, Arts/Media, Education, and Legal—clusters around roughly 28–35% frequent and 26–31% occasional use. Within that tier, Arts/Media stands out for a comparatively high occasional-use share relative to frequent use, suggesting wider experimentation without the same depth of routine integration seen in Life Science or Business/Finance. At the lower end of the distribution, Community/Social and Office/Admin occupations sit in the low-20s on frequent use, with occasional-use shares similarly moderate. Personal Care and Transport/Moving are the laggards, with frequent-use rates below 15% and occasional use in the high single digits. Overall, the cross-section is consistent with an exposure-based interpretation: occupations centered on analytical, text-heavy, and information-processing tasks concentrate frequent AI use, while frontline and service roles adopt more slowly.

Panel B shows that diffusion across the late-2025 window is occurring primarily through intensification rather than widening. Growth in frequent use is positive in most categories and is largest in Life/Physical/Social Science (roughly +10 pp), followed by Computer/Math and Business/Finance at around +7–8 pp, and Office/Admin, Legal, and Education each near +5–6 pp. By contrast, occasional use is flat or declining in nearly every occupation, with Arts/Media showing the sharpest contraction (roughly −6 pp) alongside a substantial gain in frequent use—consistent with a reallocation from experimentation toward embedded, routinized adoption. Similar though smaller offsetting patterns appear in Office/Admin, Legal, and Education. Community/Social is a notable exception, with near-zero growth on both margins. Personal Care shows a modest decline in frequent use alongside a small increase in occasional use, suggesting that in the lowest-adoption occupations some initial trial use is emerging even as regular integration has not yet taken hold.

Figure A.3 plots the evolution of the exposure–adoption gradient over time, relative to a

2023 base period. Panel A shows that the displacement exposure gradient rises monotonically from zero in 2023 to roughly 0.07 by Q1:2026, indicating that occupations with higher task-level exposure to AI displacement are adopting frequent use at an accelerating rate relative to less-exposed occupations. Panel B shows a qualitatively similar but less precisely estimated pattern for augmentation exposure, with the gradient rising from zero to roughly 0.05 by Q1:2026. The stronger displacement gradient has an intuitive interpretation: displacement exposure captures occupations where AI can directly substitute for existing tasks, creating immediate adoption pressure, whereas augmentation exposure reflects complementarity that raises the ceiling on worker output without the same urgency to integrate the tool into daily workflows.

5 Empirical Strategy

5.1 Determinants of AI Adoption

To study correlates of AI adoption, I estimate regressions of the form:

$$AI_{it} = \zeta_1\, ClearAI_{it} + \zeta_2\, Trust_{it} + \zeta_3 W_{it} + \psi\, FEAR_{it} + \beta X_{it} + \lambda_t + \varepsilon_{it}, \tag{6}$$

where AI_{it} denotes whether individual i uses AI frequently for work (e.g., daily or a few times per week; normalized to the low/never-use group), $ClearAI_{it}$ is an indicator for whether the respondent reports that the organization has communicated a clear AI strategy for integrating AI into business practices, $Trust_{it}$ is a standardized index of trust in leadership, and W_{it} is a vector of workplace practice characteristics (measured as standardized z-scores of 1–5 items, with higher values denoting better practices). The term $FEAR_{it}$ denotes an indicator for whether the respondent believes AI will replace their job within five years, and X_{it} includes demographic and job characteristics. I include wave fixed effects, λ_t, to address aggregate changes, weight observations using the survey weights, and cluster standard errors at the individual level.

Equation (6) is descriptive: it is designed to quantify whether clear organizational direction, leadership credibility, and broader workplace practices are associated with higher adoption, above and beyond worker characteristics and AI-related anxiety. A central motivation is that experimentation with AI may be constrained not only by skills and perceived returns, but also by perceived organizational guardrails and the broader management environment. In particular, a clear AI strategy may reduce ambiguity about permissible use-cases and expected conduct, while trust in leadership and supportive workplace practices may shape whether employees view experimentation as career-safe and worth the effort required to integrate AI into day-to-day workflows.

5.2 Identifying Organizational Complementarities

The main empirical results study how AI utilization and organizational conditions interact in predicting employee outcomes. The baseline panel specification is:

$$y_{it} = \sum_{k\in\{\text{Often,Sometimes}\}} \gamma_k \, \mathbb{1}\{AI_{it} = k\} + \theta \, ClearAI_{it} + \sum_{k\in\{\text{Often,Sometimes}\}} \xi_k \Big(\mathbb{1}\{AI_{it} = k\} \times ClearAI_{it} \Big) + \beta X_{it} + \alpha_i + \lambda_t + \varepsilon_{it}. \tag{7}$$

where y_{it} denotes the standardized outcome of interest (employee engagement, burnout, or job satisfaction), $\mathbb{1}\{AI_{it} = k\}$ are indicators for AI use frequency relative to the low/never-use reference group, $ClearAI_{it}$ is the clear-strategy indicator, X_{it} includes time-varying controls, α_i are person fixed effects, and λ_t are wave fixed effects. The coefficients of interest are the interaction terms ξ_k, which capture the complementarity hypothesis: whether the marginal association between AI use and outcomes is more favorable when organizations communicate a clear AI strategy.

5.3 Causal Inference and Interpretation

A key challenge is that neither AI use nor perceived organizational conditions are randomly assigned. Three threats are especially challenging to overcome. First, there are selection effects. Individuals who adopt AI may differ systematically in ambition, task complexity at work, openness to change, or baseline optimism, and these traits could also correlate with how they rate their organization. Person fixed effects, α_i, address all time-invariant worker differences, exploiting within-person changes in AI utilization and perceived organizational context.

Second, there is the potential for reverse causality and simultaneity. Engagement could itself increase adoption (i.e., more motivated workers seek out new tools), and organizations may communicate AI strategy in response to adoption waves, performance pressures, or internal events that also affect engagement and burnout. Person and wave fixed effects reduce these concerns but do not eliminate them when shocks are time-varying and jointly influence outcomes, AI use, and the communication of strategy. The heterogeneity analyses are helpful here.

Third, there are time-varying confounding macro and organizational level shocks. Aggregate shocks (e.g., industry cycles) or differential trends across groups could drive both AI adoption and outcomes. To address this, I implement specifications that flexibly control for heterogeneous macro dynamics by adding SOC3 group $\times$ wave fixed effects, absorbing differential period shocks across income strata. I also experiment with industry × wave fixed effects.

In addition to the baseline specification containing person and wave fixed effects, I also re-estimate (7) on the sample of job switchers, leveraging within-person changes around employer transitions, specifically two years before and two years after. This design increases plausibility that changes in exposure to organizational practices are causal because they isolate variation within the same individual exposed to different organizational ecosystems. While job changes are not random and sorting into higher-quality firms remains a concern, the switchers results are useful because they isolate variation in organizational environment that is mechanically linked to changes in employer. Nonetheless, the empirical strategy is not a claim of causal identification from randomized exposure. Rather, it is a disciplined effort to (i) document robust complementarities between AI use and clear organizational strategy across increasingly demanding specifications, and (ii) evaluate whether these patterns survive within-person comparisons, flexible controls for macro heterogeneity, and variation induced by employer transitions.

6 Determinants of AI Adoption

Table 1 estimates correlates of using AI frequently at work. Columns 1 through 6 enter each workplace practice separately; sample size varies because only subsets of respondents were asked particular workplace items.[7] In the single-item models, each practice is positively associated with frequent use, and all are precisely estimated: well-being (0.022, s.e. 0.003), connectedness (0.026, 0.003), feedback (0.037, 0.003), customer focus (0.013, 0.003), trust (0.019, 0.003), and respect (0.027, 0.003). Column 7 adds the clear AI strategy indicator, which is strongly predictive

[7]These differences reflect survey design (item routing) rather than non-response.

of adoption: reporting a clear strategy is associated with a 0.291 increase (s.e. 0.007) in the probability of using AI frequently, conditional on demographics and wave fixed effects.

Column 8 includes all workplace practices jointly alongside the clear-strategy indicator. The key result is that clear AI strategy remains large and precisely estimated (0.268, s.e. 0.012), while most individual workplace-practice coefficients attenuate sharply and become statistically indistinguishable from zero. The one exception is trust in leadership, which retains a positive and statistically significant coefficient (0.016, s.e. 0.007), suggesting it carries some independent predictive content beyond its covariance with strategic clarity. This pattern suggests that, in the cross section, clear AI strategy captures the dominant organization-level correlate of frequent AI use, and that the bivariate associations between frequent use and broader workplace perceptions largely reflect common covariance with strategic clarity. The column (8) coefficient implies that workers in clear-strategy organizations are about 27 percentage points more likely to report frequent AI use, holding constant demographics and wave fixed effects.[8]

Age is negatively associated with frequent use (about −0.002 per year). Education loads positively and strongly: "college or more" is associated with roughly 9–16 percentage points higher frequent use depending on the specification, and remains sizable in the saturated model (0.090). Full-time status is also strongly positive (about 5–8pp), while men report higher frequent use (about 2–3pp). White respondents report lower frequent use (roughly 3–4pp), and being married or having children is associated with modestly higher adoption. R-squared values are about 0.06–0.07 in the single-item models, rising to 0.14 when adding clear strategy (col. 7), and 0.13 in the saturated model (col. 8); sample sizes range from about 20,000 to 71,000 depending

[8]Does clear AI strategy just proxy for trust in leadership? Even when introducing person and wave fixed effects, a regression of an indicator for having a clear AI strategy on a z-score for trust in leadership produces a coefficient of .041 (p-value = 0.000).

on item availability; Figure A.3 replicates these results holding fixed the sample.[9]

Table A.5 in the Online Appendix also includes an indicator for whether an individual believes that AI is "very likely" or "somewhat likely" to displace their job in the next five years. Individuals who are at least somewhat concerned that AI will displace their job are 9–15 percentage points more likely to use AI frequently, with the coefficient stable across single-practice specifications (cols. 1–6) and attenuating modestly but remaining large and precisely estimated once clear AI strategy is added (cols. 7–8). This could indicate that frequent AI users understand that their jobs—and jobs in the economy more broadly—are subject to evolution and that the nature of their work will simply change, but not go away. Consistent with this interpretation, **?** shows that while frequent AI use raises perceived displacement risk, this increase is substantially offset among workers in high-quality workplace environments, suggesting that managerial practices shape not only AI adoption but also how workers interpret its consequences for their own employment.

Are these coefficients biased due to having too precise of a cutoff on AI utilization? Table A.6 in the Online Appendix replicates these results using the full distribution of AI utilization on a 0-10 scale: Daily=10, Few Times Week=8, Few Times Month=6, Few times Year=4, Once a Year=2, Less than Once a Year=1, Never/Other=0. Results are largely similar.

Table 2 subsequently explores the correlates of whether the respondent reports having a clear AI strategy, a yes/no response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Columns (1)–(6) enter each workplace practice

[9]The coefficient estimates are broadly similar across the two tables, confirming that the main results are not driven by compositional variation in the sample. A few point estimates shift modestly: the feedback coefficient is 0.037 in Table 1 versus 0.029 in Table A.3; trust rises from 0.019 to 0.028; respect falls from 0.027 to 0.022. The clear AI strategy coefficient in column (7) is 0.291 in the varying-sample specification versus 0.274 in the stable sample. Column (8) estimates are nearly identical across both tables. Demographic coefficients shift modestly across the single-item columns but converge in the saturated specification.

separately and show that perceived management quality is strongly associated with strategic clarity. The largest single-item associations are for feedback (0.060, s.e. 0.003), connectedness (0.051, 0.003), and well-being (0.050, 0.003), with trust (0.043, 0.004), respect (0.042, 0.003), and customer focus (0.040, 0.003) also positive and precisely estimated. Interpreted as linear probability estimates, a one standard deviation increase in feedback or connectedness is associated with roughly a 5–6 percentage point higher probability of reporting a clear AI strategy.[10]

Column (7) includes all workplace practices jointly. Feedback remains the most robust correlate of clear strategy (0.040, s.e. 0.006), suggesting that communication and performance dialogue are tightly linked to whether employees perceive their organization as having an articulated AI plan. The coefficients on connectedness and trust remain positive but attenuate (0.017 and 0.016, respectively), while well-being, customer focus, and respect become negligible and statistically indistinguishable from zero. The saturated specification indicates that clear AI strategy is most closely linked to organizational communication and developmental feedback, with broader positive workplace perceptions correlating with strategy primarily through shared covariance with those communication channels. The main results in the next section will show, the complementarities in worker outcomes are concentrated in trust rather than feedback or connectedness.

The demographic gradients are stable across columns. Education is strongly predictive: "college or more" is associated with about 10 percentage points higher reported strategy clarity, and full-time workers are about 5–9 percentage points more likely to report a clear strategy. Age loads negatively (roughly -0.002 per year), White respondents are less likely to report clear strategy (about 3–4pp), and Black respondents are more likely (about 2–4pp). These patterns suggest that reported strategic clarity is not purely an organization-level object; it also reflects heterogeneous

[10] Table A.4 replicates these results using a stable sample only on observations with non-missing culture variables.

exposure to communication and role context within firms.

7 Organizational Complementarities of AI Adoption

7.1 Main Results

Table 3 examines whether the relationship between employees' AI utilization and workplace outcomes depends on whether the organization has communicated a clear AI strategy. The baseline specification regresses standardized outcomes—employee engagement (on four of the Q12), burnout, and job satisfaction—on indicators for AI use frequency (frequent and occasional, relative to a low/never-use reference group), an indicator for whether the organization has articulated a clear AI strategy, and their interactions, alongside demographic controls and person and wave fixed effects. The interaction terms capture complementarities: whether the marginal association of AI use with outcomes is larger (or smaller) when a clear strategy is present.[11]

Across outcomes, the pooled baseline estimates with demographics (cols. (1), (5), and (9)) show a consistent pattern of complementarity. For employee engagement, AI use in isolation is weakly related to engagement, while the interaction with clear strategy is economically large and precisely estimated. In particular, frequent AI use has a near-zero main effect (-0.014 SD), whereas the interaction with clear strategy is 0.213 SD, implying that frequent users in organizations with a clear strategy exhibit roughly 0.199 SD higher engagement relative to the low/never-use group. Occasional use is positively associated with engagement (0.045 SD), and the corresponding interaction adds an additional 0.068 SD, yielding an implied association of approximately 0.113 SD

[11] Figure A.6 in the Online Appendix provides a visual representation of these results. I use a sub-sample of the Q12 due to missing values that limit the sample size, which pushes the data too hard when restricting the sample to switchers.

when a clear strategy is present. For burnout, the same complementarity appears with the opposite sign. Frequent AI use is associated with higher burnout (0.115 SD), but the interaction with clear strategy is negative (-0.074 SD), offsetting much of the increase. For occasional use, neither the main effect (0.030 SD) nor the interaction (-0.039 SD) is precisely estimated, suggesting the burnout complementarity is concentrated among the most intensive users. Job satisfaction mirrors the engagement results. Frequent AI use has a negative main effect (-0.054 SD), but the interaction with clear strategy is strongly positive (0.144 SD), implying an overall association of roughly 0.090 SD for frequent users when a clear strategy is present. The interaction for occasional use is smaller (0.037 SD) and imprecisely estimated. Cumulatively, the baseline estimates indicate that the association between AI utilization and employee outcomes is more favorable in organizations that communicate a clear AI strategy, strengthening engagement and satisfaction while attenuating the burnout penalty associated with AI use absent such strategy.

Recognizing the potential for self-selection among more ambitious and productive individuals into using AI, columns (2), (6), and (10) introduce person and wave fixed effects, absorbing all time-invariant worker heterogeneity and common period shocks. Once these fixed effects are included, the level association of having a clear AI strategy becomes small across outcomes (0.046 for engagement, -0.003 for burnout, and 0.050 for satisfaction), though it retains marginal significance for engagement and satisfaction. The interaction terms attenuate relative to the pooled estimates but remain positive for engagement: the coefficient on frequent AI use interacted with clear strategy is 0.064 SD and the corresponding coefficient for occasional use is 0.051 SD (col. (2)), implying meaningful engagement gains among AI users when a clear strategy is present. For burnout, the interactions become small and statistically indistinguishable from zero (col. (6)), suggesting the burnout mitigation result is partly driven by cross-sectional sorting. For job satisfaction, the interaction terms also attenuate and are no longer statistically distinguishable from zero (col. (10)), though they remain positive in sign.

Columns (3), (7), and (11) augment the person fixed effects specification with three-digit occupation by wave fixed effects, flexibly controlling for differential trends and macro shocks across occupational groups, which addresses concerns about time-varying skill premia or AI advances that benefit one occupation more than another. The engagement complementarities strengthen relative to the person-and-time-FE specification: 0.071 SD for frequent AI use interacted with clear strategy and 0.083 SD for occasional use interacted with clear strategy (col. (3)), with the latter now precisely estimated. For burnout and job satisfaction, the interactions remain small and statistically indistinguishable from zero (cols. (7) and (11)), indicating that the primary channel through which a clear AI strategy improves outcomes runs through engagement rather than burnout relief or satisfaction—at least when exploiting within-person variation.

Because the Gallup Panel does not include a direct employer identifier, I infer job switches from within-person resets in self-reported tenure, classifying a transition as an employer change whenever tenure drops by more than one year between adjacent waves. The one-year threshold is chosen from the empirical distribution of negative tenure changes (Figure A.4): drops below one year are concentrated near zero and plausibly reflect rounding or within-employer recoding, while the cumulative distribution (Figure A.5) confirms that this cutoff excludes roughly 40 percent of negative changes—the noisiest region—while retaining the bulk of plausible transitions. Results are robust to thresholds of 0.5 and 2 years. Under this definition, 93.2 percent of switchers experience exactly one transition during the panel. I anchor on the first observed switch and restrict the estimation window to ± 2 waves. The clear-AI-strategy indicator in the switcher specifications is measured contemporaneously at each wave, reflecting the respondent's perception of their current employer's organizational context. Because switchers change employers within the event window, this coding allows the strategy variable to capture the shift in organizational environment that accompanies the transition. Person fixed effects absorb all time-invariant worker characteristics, so the main effect of clear AI strategy and its interactions with AI use are identified from within-person changes in perceived organizational context around the employer transition—precisely the variation this design is intended to isolate. Missingness in the strategy variable is low and modestly higher post-switch (5–7% pre versus 11–12% post). The resulting switcher sample comprises 9,841 person-wave observations, of which roughly 23 percent report a clear AI strategy—a share that is stable across event-time periods.

Columns (4), (8), and (12) restrict the sample to job switchers and estimate the model using within-person variation around employer transitions, exploiting a potentially more exogenous source of variation in organizational environment. For engagement, the interaction coefficient for frequent use is 0.153 SD (col. (4)), directionally consistent with the pooled and fixed-effects results though imprecisely estimated given the smaller sample; the occasional-use interaction is near zero (-0.005). For burnout, the complementarity is stronger than in the full panel: the frequent-use main effect is 0.272 SD—indicating that switchers who adopt AI intensively without clear organizational guidance experience markedly elevated burnout—while the interaction with clear strategy is -0.234 SD (col. (8)), offsetting most of the penalty. For job satisfaction, frequent AI use carries a large negative main effect among switchers (-0.235 SD), but the interaction with clear strategy is positive (0.220 SD), again largely neutralizing the adverse association (col. (12)). The switcher estimates are noisier given the smaller sample (roughly 3,500–3,800 observations), but they are qualitatively consistent with the full-panel results and, for burnout and satisfaction.

Figure A.7 plots predicted engagement and job satisfaction across the full AI utilization frequency distribution, separately for workers in organizations with and without a clear AI strategy. For engagement (Panel A), predicted values are flat and near zero across all frequency levels absent a clear strategy, whereas they rise monotonically from roughly 0.07 SD at the "never" margin to over 0.30 SD at daily use when a clear strategy is present—implying that the engagement returns to AI use accrue almost entirely in organizations that provide strategic direction. Job satisfaction (Panel B) exhibits an analogous but more striking pattern: without a clear strategy, predicted satisfaction *declines* with AI use frequency, falling from roughly −0.05 SD among non-users to approximately −0.10 SD among daily users, while the clear-strategy group rises from about 0.10 SD to 0.24 SD. The widening gap between the two groups at higher frequency levels reinforces the complementarity results: intensive AI use is associated with worse outcomes in organizations that lack clear guidance and with substantially better outcomes in those that provide it.

To understand the dimensions of engagement that are influenced more by these complementarities, Table 4 replicates the baseline specification with each of the 12 engagement components. First, the level association of having a clear AI strategy is generally small across items, with statistically detectable gains concentrated in recognition and mission-oriented dimensions: the strategy indicator is positive for "Do Best" (0.051, col. (3)), "Recognition" (0.072, col. (4)), "Opinions" (0.050, col. (7)), and "Mission" (0.093, col. (8)), while remaining imprecisely estimated for most other components. Second, the interaction terms indicate that the organizational strategy environment conditions how AI use maps into specific engagement margins. Absent the interaction, frequent AI use is negatively associated with access to materials: the coefficient is −0.093 for "Materials" (col. (2)), while remaining near zero or weakly negative for most other items.

The interaction terms show where complementarities are strongest. For "Materials" (col. (2)), the interaction is large and precisely estimated for frequent users: 0.150 for "Often × Clear AI

Strategy." This estimate implies that a clear strategy more than offsets the negative main effect of frequent AI use on perceived resourcing ($-0.093 + 0.150 \approx 0.057$). For "Development" (col. (6)), the interaction for frequent users is 0.081 and precisely estimated, indicating that AI use in clear-strategy organizations is more strongly associated with employees feeling that someone encourages their growth. "Do Best" (col. (3)) exhibits a similar pattern: the interaction for occasional users is 0.070 and precisely estimated, while the interaction for frequent users is positive (0.036) but imprecise. Complementarities also appear for "Best Friend" (col. (10)), where the interaction for occasional use is 0.080 and precisely estimated, and the interaction for frequent use is positive (0.043) but not statistically distinguishable from zero. For "Progress" (col. (11)), the interaction for frequent users is 0.081 and marginally significant. Clear AI strategy most reliably amplifies the developmental and relational dimensions of engagement—the margins most likely to reflect whether employees view AI experimentation as organizationally supported and career-enhancing.

Table 5 re-estimates the engagement baseline specification separately by demographic subgroup. The level effect of reporting a clear AI strategy is small in most subgroups, though it is positive and precisely estimated among women (0.079, s.e. 0.031), college-educated workers (0.041, s.e. 0.019), and White respondents (0.058, s.e. 0.026). Complementarities are most pronounced among men and among respondents with at least a college education, which is intuitive given exposure among college graduates to AI (Eloundou et al., 2024). For men, the interaction for frequent use is 0.079 (s.e. 0.038) and precisely estimated, while the main effect of frequent AI use is near zero (0.016), implying that the engagement return to intensive AI use accrues primarily when paired with clear organizational strategy (implied effect of roughly 0.095 SD). For women, the interaction terms are positive (0.049 for frequent and 0.055 for occasional) but imprecisely estimated given the smaller effective sample of female frequent users. Among respondents with at least a college education, the interaction for frequent use is 0.061 (s.e. 0.026) and precisely estimated, with a near-zero main effect (0.012), yielding an implied effect of approximately 0.073 SD. Among respondents without a college degree, the interaction is larger in magnitude (0.085) but imprecisely estimated (s.e. 0.057), making it difficult to rule out smaller complementarities.

Among White respondents, the complementarity for occasional use is positive and marginally significant (0.065, s.e. 0.036), while the interaction for frequent use is positive (0.027) but imprecise. Among non-White respondents, the interaction for frequent use is notably large (0.134, s.e. 0.053) and precisely estimated—the strongest subgroup complementarity in the table—despite the smaller sample. The main effect of frequent AI use among non-White workers is near zero (0.009), so the implied engagement gain for frequent users in clear-strategy organizations is roughly 0.143 SD. This result suggests that the organizational enablement captured by a clear AI strategy may be especially consequential for non-White workers, potentially because strategic clarity reduces ambiguity in settings where baseline trust or inclusion may be lower. The occasional-use interactions are small and imprecise across both racial subgroups.

7.2 Credibility, Psychological Safety, and Realized Gains

Table 3 shows that AI use is not uniformly associated with better outcomes; instead, the association becomes sharply more favorable when the organization communicates a clear AI strategy, raising engagement and satisfaction and largely offsetting the burnout penalty that appears for adopters in low-clarity environments. A natural mechanism is psychological safety: strategy reduces ambiguity about purpose and guardrails, but employees still need to believe that experimentation is career-safe and that leadership will interpret early mistakes as learning rather than misuse.

Table 6 indicates that workplace trust is a first-order correlate of employee well-being, and that it meaningfully conditions how AI adoption maps into engagement in several specifications. Across outcomes, the trust index enters with the expected signs and is large in magnitude: higher trust is strongly associated with higher engagement (0.658 SD in the pooled specification, 0.419 SD with person and time FE) and job satisfaction (0.649 and 0.480 SD, respectively), and with lower burnout (−0.406 and −0.281 SD). The trust coefficients remain sizable among job switchers (0.507 for engagement, −0.414 for burnout, 0.591 for satisfaction), suggesting that changes in perceived trust track changes in these outcomes within workers over time and across employers.

The interaction terms provide a more rigorous test of the psychological-safety mechanism. For employee engagement, the AI-use-by-trust interactions are small in the pooled-with-demographics specification (col. (1)), but become positive once person and wave fixed effects are introduced. In particular, the interaction for frequent users is 0.033 in col. (2) and rises to 0.054 in the SOC3 $\times$ time specification (col. (3)), where it is marginally significant. Holding constant worker fixed effects and occupation-specific trends, AI use is more strongly associated with higher engagement in higher-trust workplaces. Trust appears to amplify the engagement returns to AI adoption, consistent with the idea that employees are more willing to experiment, share failures, and integrate AI into workflows when leadership is perceived as fair and supportive. For occasional users, the trust interaction is small in the person-FE specification (0.024 in col. (2)) but positive and significant in the pooled burnout model (0.036 in col. (5)). In the person-FE burnout specification, however, the occasional-use interaction turns negative and is imprecisely estimated (−0.014, col. (6)), suggesting that the trust-mediated attenuation of the burnout penalty for lighter AI users is driven primarily by cross-sectional variation rather than within-person changes.

For burnout, the interaction terms for frequent users are generally small and statistically indistinguishable from zero across the fixed-effects specifications (cols. (6)–(7)), indicating limited evidence that trust systematically moderates the burnout penalty of intensive AI use. For job satisfaction, the pooled specification shows a positive and significant interaction for frequent users (0.039 in col. (9)), and the SOC3 $\times$ time specification yields the largest point estimate (0.049 in col. (11)), though the latter is only marginally significant. The interaction estimates attenuate and lose precision in the person-FE specification without occupation trends (col. (10)). The switcher estimates are noisier due to the substantially smaller samples (765–873 observations) and do not deliver robust moderation effects for any outcome.

In this sense, Table 6 complements Table 3 by showing that psychological safety, proxied here by workplace trust, is tightly linked to baseline levels of engagement and satisfaction and is plausibly an enabling condition for translating AI use into higher engagement once individual heterogeneity is absorbed. The moderation evidence is strongest for engagement—particularly in specifications that control for occupation-specific trends—and weaker for burnout and job satisfaction, suggesting that trust primarily affects whether AI adoption feels motivating and empowering, rather than directly offsetting burnout in the way a clear strategy appears to do.

Table 7 examines a candidate mechanism by linking workplace trust to employees' perceived productivity gains from genAI. In pooled cross sections, frequent users report substantially higher perceived AI productivity: 1.111 s.d. (s.e. $=0.233$) for the z-score outcome and 0.230 (s.e. $=0.036$) for the "5/5" indicator in cols. 1 and 4. Introducing person fixed effects and time controls attenuates these main effects in cols. 2–3 and 5–6 (e.g., 0.183 and 0.388 for the z-score outcome; -0.038 and -0.129 for the binary outcome), consistent with selection into frequent use. Occasional users show a similar pattern: the pooled z-score effect is 0.524 (s.e. $=0.234$) and attenuates to near zero once person fixed effects are introduced.

By contrast, interactions with workplace trust are large and strengthen for frequent users once individual heterogeneity is absorbed. For the z-score outcome, the "frequent" × trust coefficient is small in the pooled model (0.037, s.e. =0.202) but rises sharply to 0.610 (s.e. =0.270) with person and time FE and to 0.718 (s.e. =0.251) with SOC3 × time FE in cols. 2–3. For the "5/5" indicator, the corresponding interactions are 0.095 (s.e. =0.029), 0.392 (s.e. =0.128), and 0.408 (s.e. =0.127) in cols. 4–6—all precisely estimated. The pattern implies that in the pooled cross section, the trust–productivity link for frequent users is obscured by compositional differences, but within-person variation reveals a strong complementarity: workers who use AI frequently report substantially larger perceived productivity gains when they also experience higher trust in leadership. Interactions for "occasional" users are smaller and less robust in the z-score specifications—0.027 (0.202), 0.438 (0.365), and 0.246 (0.203) in cols. 1–3—but the binary outcome tells a cleaner story: the interaction is 0.044 (s.e. =0.024) in the pooled model, 0.293 (s.e. =0.124) with person and time FE, and 0.337 (s.e. =0.098) with SOC3 × time FE, all positive and the latter two precisely estimated. The evidence suggests that trust amplifies perceived productivity returns for both frequent and occasional users, with the effect most reliably detected for frequent users in the z-score metric and for both groups in the binary indicator.

Consistent with this hypothesis of psychological safety, better communication and trust should also shift how employees use AI—not just whether they use it. After restricting the sample to those who use AI at least sometimes, Figure 7 examines how the composition of AI use-cases varies based on whether respondents report their firm having a clear AI strategy, or whether they report high trust in senior leadership. In Panel A, respondents who report that their firm has communicated a clear AI strategy are more likely to use AI across a broad set of concrete work tasks. The largest absolute gaps appear for consolidating information (roughly 45% versus 38%), identifying problems (25% versus 15%), and making predictions (14% versus 7%)—use cases that involve synthesizing or structuring information within workflows. Clear-strategy adopters also report higher shares for automating basic tasks (42% versus 33%), learning new things (35% versus 31%), interacting with customers (17% versus 11%), collaborating with others (15% versus 8%), and setting up or monitoring devices (11% versus 7%). Generating ideas and consolidating information are the most prevalent overall, with shares near or above 40% in both groups, though the clear-strategy group edges ahead on consolidation. By contrast, the residual "Other" category is smaller for the clear-strategy group (roughly 8% versus 14%), consistent with strategy shifting AI use away from hard-to-classify experimentation and toward more standardized applications.

Panel B indicates a related but more selective pattern for trust in leadership. The largest gap between high- and low-trust adopters appears for learning new things: roughly 37% of high-trust adopters report this use case versus 28% of low-trust adopters, a gap of nearly 10 percentage points and the widest in the panel. Idea generation and information consolidation are the most prevalent use cases overall, with shares near or above 40% in both groups, though high-trust adopters edge modestly ahead on both (roughly 45% versus 40% for idea generation; 43% versus 42% for consolidation). High-trust adopters are also modestly higher on identifying problems (roughly 20% versus 19%). Differences in routine applications are comparatively compressed: basic task automation is nearly identical across groups (roughly 37% each), and customer interaction is similarly close. Collaboration with others is roughly comparable across trust levels (approximately 11–12% in both groups), and making predictions shows little differentiation (roughly 10% in each group). Setup/monitoring of devices is one area where low-trust adopters report slightly higher use (roughly 10% versus 8%). The residual "Other" category is marginally smaller for the high-trust group (roughly 10% versus 12%). These results are consistent with a psychological-safety mechanism in which stronger trust is associated not simply with *more* AI use, but with a tilt toward higher-value, more exploratory applications—particularly learning—among those already adopting, while routine and standardized use cases show little differentiation by trust.

8 Conclusion

How has AI adoption evolved in the United States, and when does adoption translate into better workplace outcomes? Using nationally representative longitudinal data from 2023 to Q1:2026, I document rapid diffusion of AI use at work. Frequent use rose from under 10% in 2023 to over 27% by Q1:2026, and occasional use expanded from roughly 10% to 21% over the same period. Adoption is not confined to a narrow slice of the labor market: while diffusion is fastest in analytic and information-processing roles, it is visible across sectors—including the public sector and non-profits—and throughout the income distribution, with the sharpest intensification occurring among active users from mid-2025 onward as occasional adopters converted to frequent use. Importantly, diffusion is structured by organizational hierarchy. Senior leaders report the highest intensity of use, reaching the low-40s in frequent use by Q4:2025, with supervisors and project managers close behind, consistent with an internal transmission process in which adoption spreads through managerial channels rather than arriving solely as individual experimentation.

But adoption alone does not improve employee outcomes. The pooled cross-sectional estimates are striking: frequent AI use without a clear organizational strategy carries a near-zero association with engagement (-0.014 SD) and a positive association with burnout (0.115 SD), while the interaction with clear strategy is large enough to flip both signs—implying roughly 0.20 SD higher engagement and a substantially attenuated burnout penalty for frequent users in clear-strategy environments. These complementarities survive, in attenuated form, within-person specifications that absorb all time-invariant worker heterogeneity. With person and time fixed effects, the engagement interaction for frequent users remains positive (0.064 SD) and strengthens modestly with occupation-by-time controls (0.071 SD). The burnout and satisfaction complementarities, however, attenuate in within-person specifications on the full panel, suggesting they are partly driven by cross-sectional sorting. The exception is the job-switcher sample, where variation in organizational environment is mechanically tied to employer transitions: among switchers, frequent use without clear strategy raises burnout by 0.272 SD, and the interaction with clear strategy offsets most of this penalty (−0.234 SD); a parallel pattern appears for job satisfaction (−0.235 SD main effect, 0.220 SD interaction). Across the Q12 components, the within-person complementarities concentrate in access to materials, developmental encouragement, and progress—dimensions most plausibly affected by whether the organization supports experimentation.

Trust in leadership further conditions these returns and suggests psychological safety for experimentation: within-person estimates show that frequent users in high-trust environments report 0.61–0.72 SD larger perceived productivity gains from AI, and the trust–engagement interaction strengthens in specifications that control for occupation-specific trends (0.054 SD). Trust also shifts the composition of use: high-trust adopters are substantially more likely to use AI for learning, while routine applications like basic task automation are flat across trust levels. The evidence is consistent with a psychological-safety mechanism: clear strategy reduces ambiguity about permissible use, trust reduces the perceived cost of early mistakes, and together they govern whether adoption translates into productive integration or stalls at superficial experimentation.

These findings have practical implications for firms and public-sector organizations seeking AI-enabled productivity gains. Investments in tools and infrastructure are unlikely to be sufficient without complementary investments in managerial capacity—communication, feedback, and trust-building—that shape how employees interpret guardrails and whether they view experimentation as career-safe. Future research can extend this work by linking adoption patterns more directly to objective performance measures, tracing the distinct role of middle management in scaling use cases, and studying how governance choices (training, monitoring, incentives, and risk frameworks) affect both productivity and worker well-being over longer horizons.

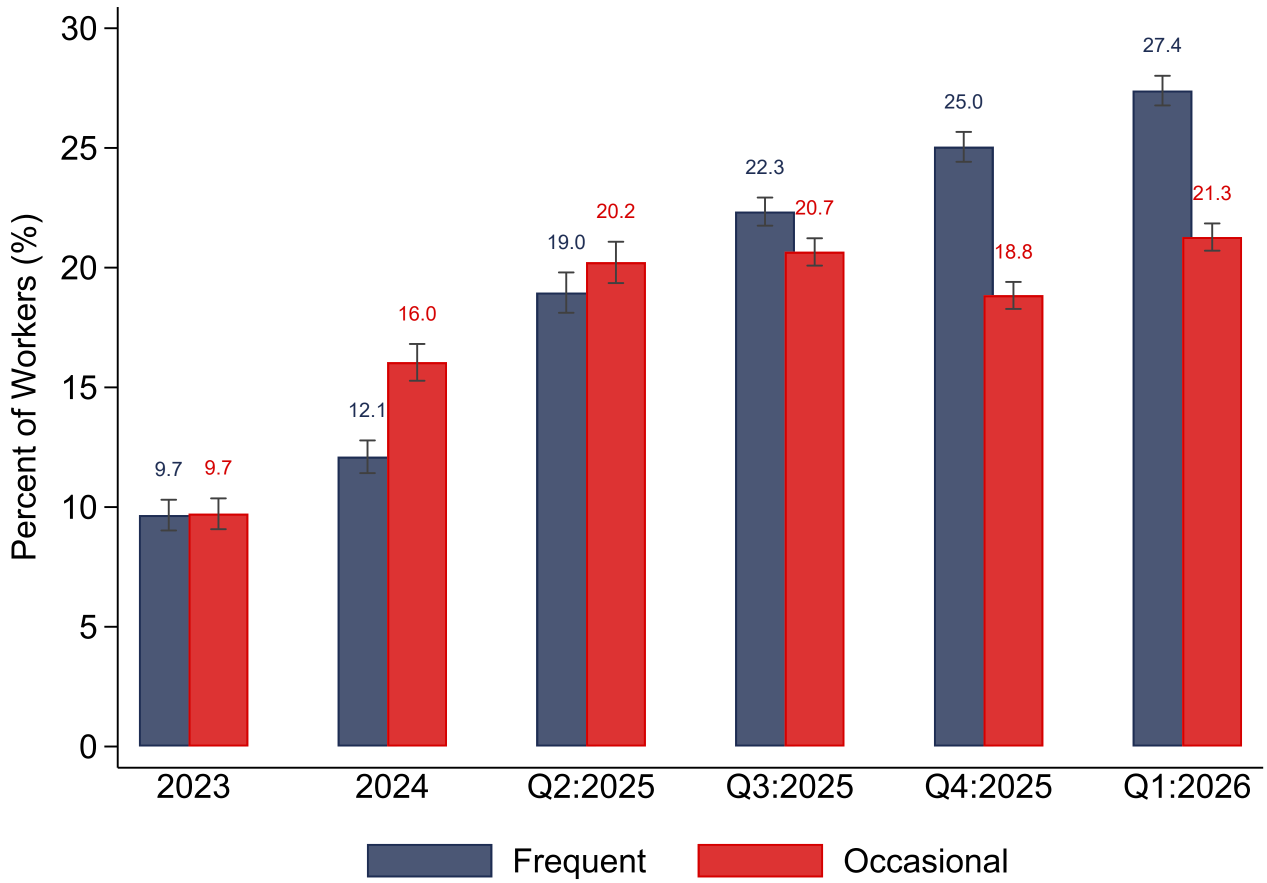


Figure 1: Aggregate AI Utilization Over Time

Notes.—Sources: Gallup Panel, 2023–2026. The figure plots the share of respondents who answer the question "How often do you use artificial intelligence in your role," which include: (a) daily, (b) a few times a week, (c) a few times a month, (d) a few times a year, (e) once a year, (f) less often than once a year, and (g) never. I consolidate AI use into two categories: frequent (daily and a few times a week) and occasional (a few times a month, a few times a year). Before asking this question, respondents are given a definition of AI: "Artificial intelligence is when a computer does or enhances a task that is normally performed by a person." The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

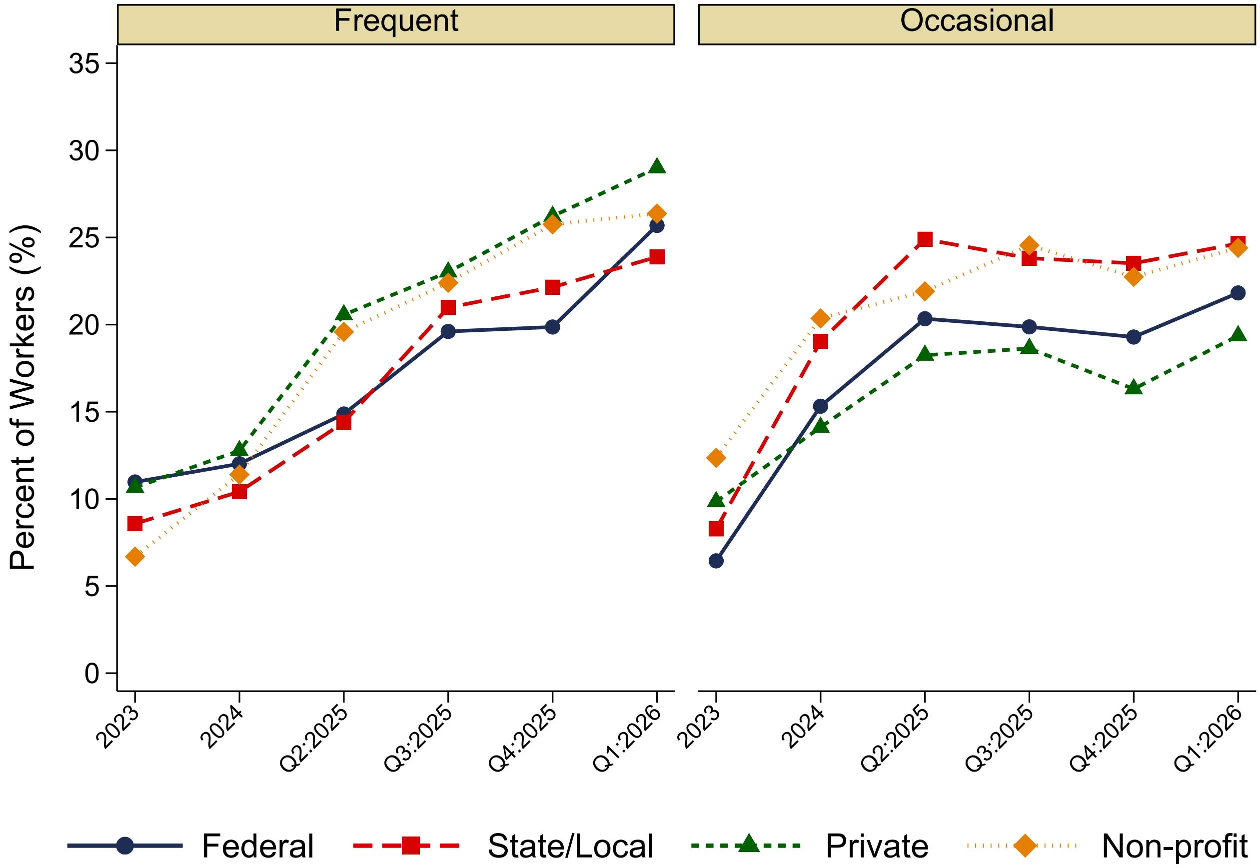


Figure 2: AI Adoption Across Public and Private Sectors Over Time

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The figure plots the share of respondents separately for federal, state/local, private, and non-profit sector employees who answer the question "How often do you use artificial intelligence in your role," which include: (a) daily, (b) a few times a week, (c) a few times a month, (d) a few times a year, (e) once a year, (f) less often than once a year, and (g) never. I consolidate these answers into two groups: frequent (daily and a few times a week) and occasional (a few times a month and a few times a year). Each panel tracks one use category over time, with separate connected lines for each employment type. Before asking this question, respondents are given a definition of AI: "Artificial intelligence is when a computer does or enhances a task that is normally performed by a person." The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

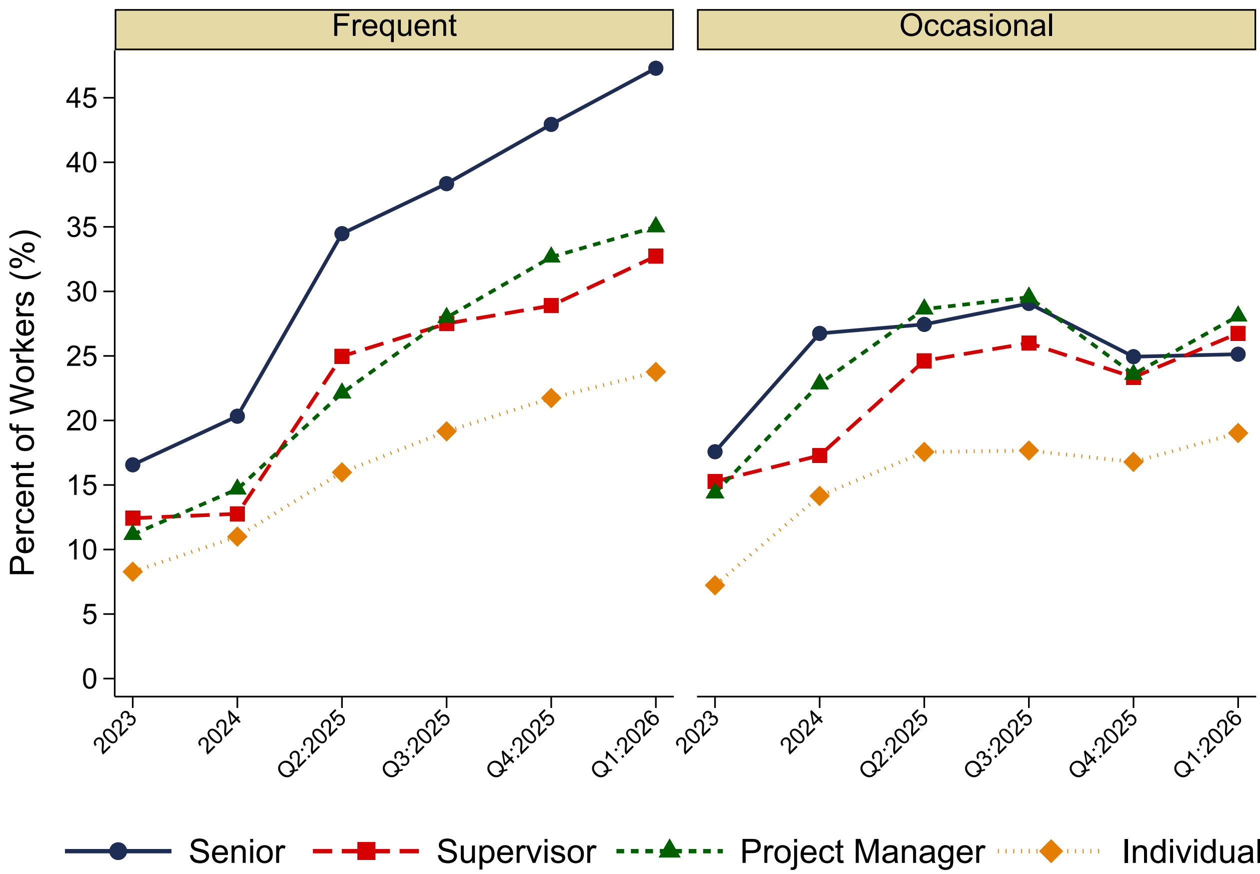


Figure 3: AI Adoption Across Employee Hierarchical Structure Over Time

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The figure plots the share of respondents separately for employees at different levels of the organizational hierarchy who answer the question "How often do you use artificial intelligence in your role," which include: (a) daily, (b) a few times a week, (c) a few times a month, (d) a few times a year, (e) once a year, (f) less often than once a year, and (g) never. I consolidate these answers into two groups: frequent (daily and a few times a week) and occasional (a few times a month and a few times a year). Each panel tracks one use category over time, with separate connected lines for each role. The hierarchy is based on the answer to the following: "Which of the following best describes your role or job responsibilities within your organization? (a) Senior leader—I am a senior leader, or a leader who manages other managers; (b) Supervisor—I am a manager or supervisor whose main responsibilities are the work output of other people; (c) Project Manager—I am a project manager who is responsible for other people's work; (d) Individual Contributor—I am responsible for my own work or output." Before asking the AI use question, respondents are given a definition of AI: "Artificial intelligence is when a computer does or enhances a task that is normally performed by a person." The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

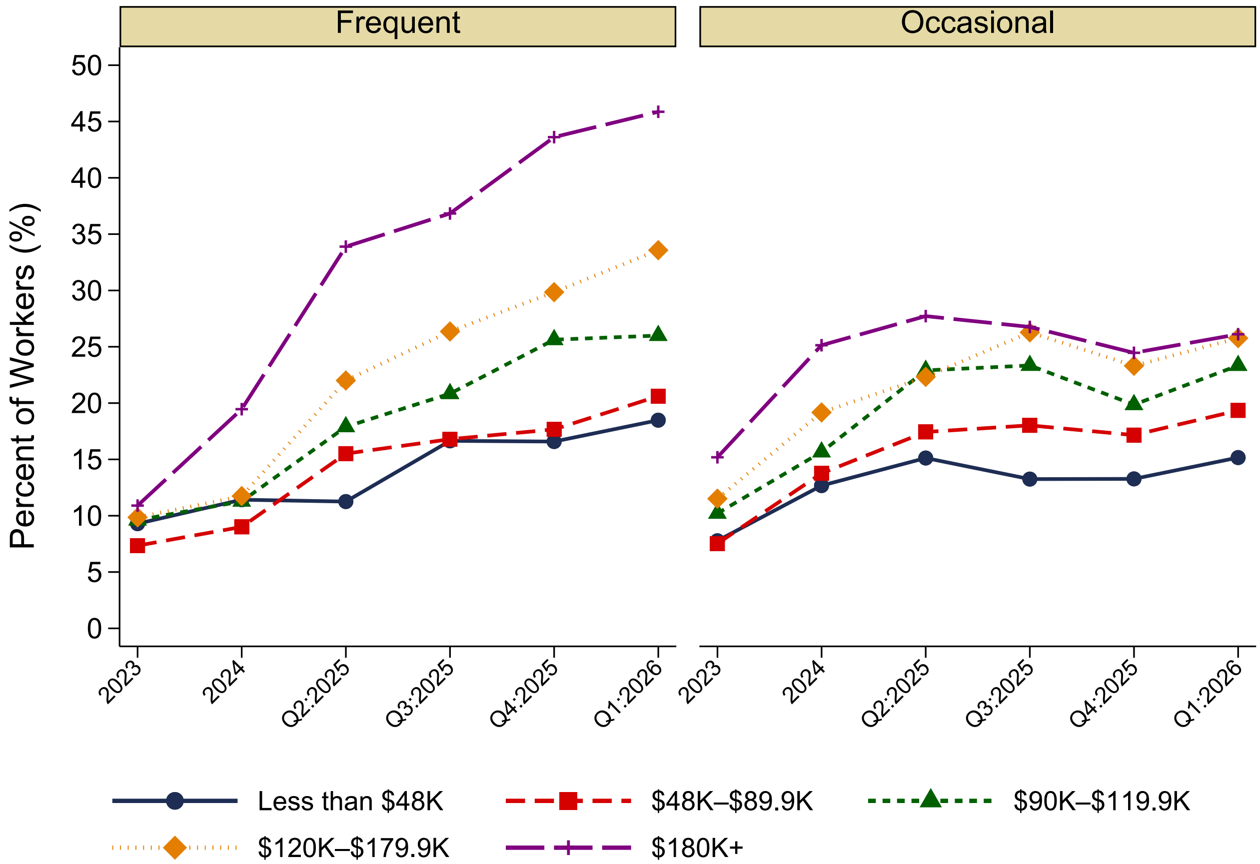
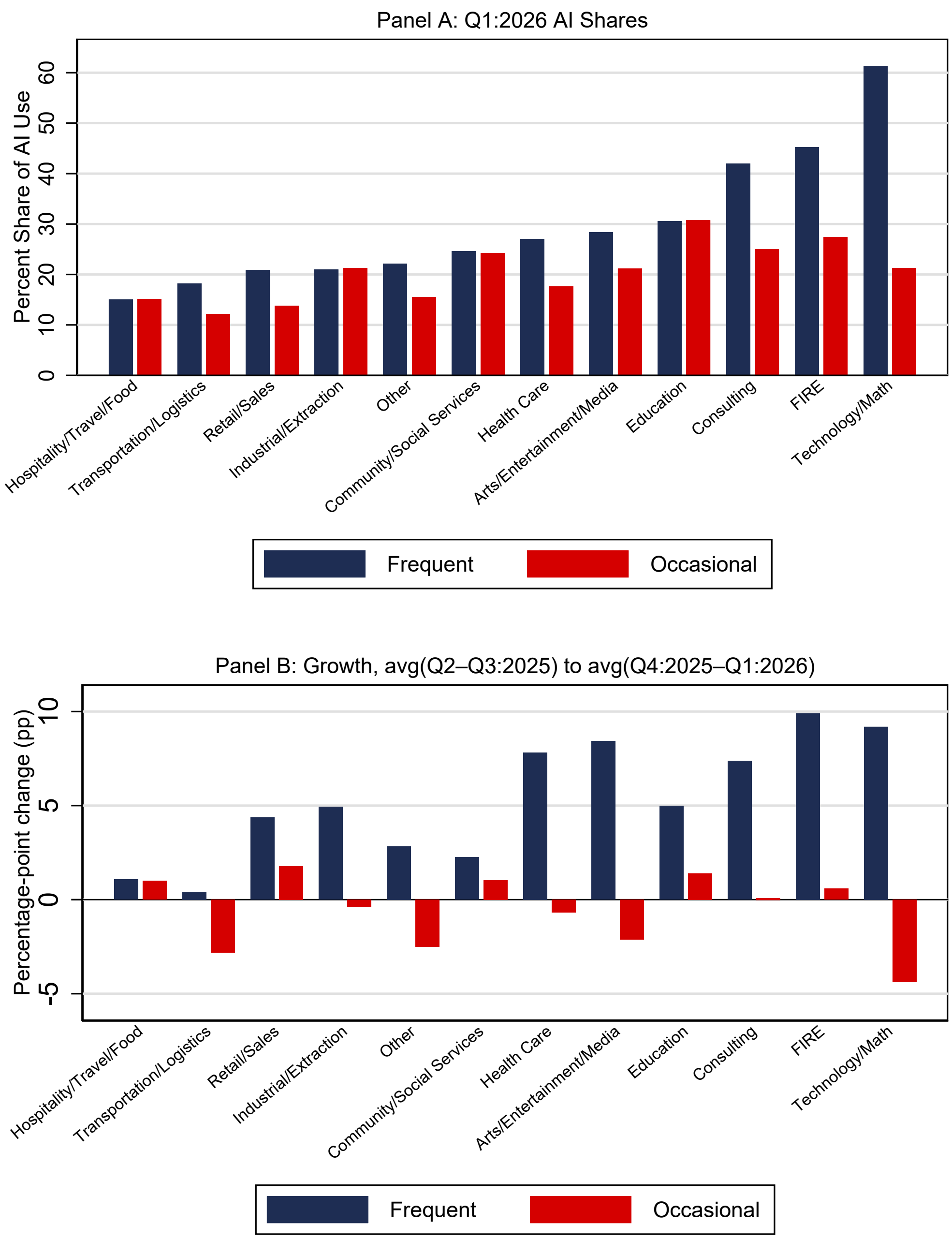


Figure 4: AI Adoption Across Household Income Over Time

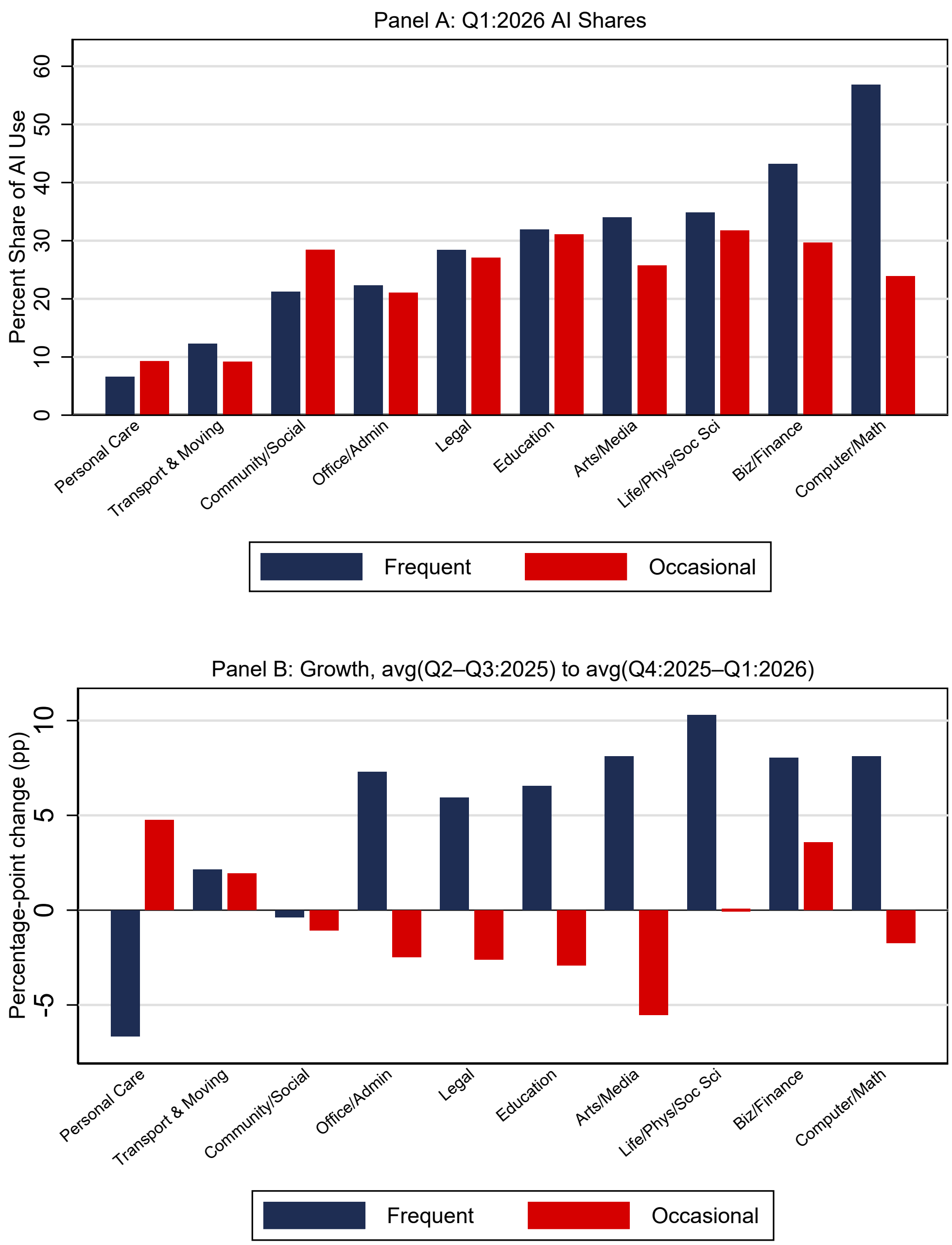
Notes.—Sources: Gallup Panel, 2023–Q1:2026. The figure plots the share of respondents separately for employees at different household income levels who answer the question How often do you use artificial intelligence in your role," which include: (a) daily, (b) a few times a week, (c) a few times a month, (d) a few times a year, (e) once a year, (f) less often than once a year, and (g) never. I consolidate these answers into two groups: frequent (daily and a few times a week) and occasional (a few times a month and a few times a year). Each panel tracks one use category over time, with separate connected lines for each income group. Income is based on the respondent's reported annual household income, collapsed into five groups: less than $48K, $48K–$89.9K, $90K–$119.9K, $120K–$179.9K, and $180K and above. Before asking the AI use question, respondents are given a definition of AI: Artificial intelligence is when a computer does or enhances a task that is normally performed by a person." The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Figure 5: Cross-sectional and Time Series Variation Across Industries



Notes.—Sources: Gallup Panel, 2023–Q1:2026. Panel A plots the share of respondents who answer the question How often do you use artificial intelligence in your role," which include: (a) daily, (b) a few times a week, (c) a few times a month, (d) a few times a year, (e) once a year, (f) less often than once a year, and (g) never. I consolidate these answers into two groups: frequent (daily and a few times a week) and occasional (a few times a month and a few times a year), separately by two-digit industry, based on the Q1:2026 wave. Before asking this question, respondents are given a definition of AI: Artificial intelligence is when a computer does or enhances a task that is normally performed by a person." Panel B plots the percentage-point change from the average of Q2:2025 and Q3:2025 to the average of Q4:2025 and Q1:2026. The sample is restricted to full- and part-time workers aged 20–65, industries with at least 300 respondents per wave, and the observations are weighted.

Figure 6: Cross-sectional and Time Series Variation Across Occupations



Notes.—Sources: Gallup Panel, 2023–Q1:2026. Panel A plots the share of respondents who answer the question How often do you use artificial intelligence in your role," which include: (a) daily, (b) a few times a week, (c) a few times a month, (d) a few times a year, (e) once a year, (f) less often than once a year, and (g) never. I consolidate these answers into two groups: frequent (daily and a few times a week) and occasional (a few times a month and a few times a year), separately by two-digit occupation, based on the Q1:2026 wave. Before asking this question, respondents are given a definition of AI: Artificial intelligence is when a computer does or enhances a task that is normally performed by a person." Panel B plots the percentage-point change from the average of Q2:2025 and Q3:2025 to the average of Q4:2025 and Q1:2026. The sample is restricted to full- and part-time workers aged 20–65, occupations with at least 300 respondents per wave, and the observations are weighted.

Table 1: Predicting High AI Utilization Using Workplace Practices

	Dep. var. = Use AI Frequently (1/0)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
C: Well-being	.022***							-.011
	[.003]							[.007]
C: Connected		.026***						.006
		[.003]						[.006]
C: Feedback			.037***					.008
			[.003]					[.005]
C: Customers				.013***				-.007
				[.003]				[.005]
C: Trust					.019***			.016**
					[.003]			[.007]
C: Respect						.027***		.005
						[.003]		[.006]
Has Clear AI Strategy							.291***	.268***
							[.007]	[.012]
Age	-.002***	-.002***	-.002***	-.002***	-.003***	-.002***	-.001***	-.002***
	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]
Some College	.031***	.022**	.039***	.020**	.009	.035***	.029***	.009
	[.008]	[.009]	[.009]	[.009]	[.010]	[.009]	[.007]	[.012]
College or More	.144***	.127***	.156***	.129***	.094***	.150***	.122***	.090***
	[.008]	[.008]	[.008]	[.008]	[.010]	[.008]	[.007]	[.012]
Male	.027***	.026***	.023***	.025***	.025***	.023***	.025***	.013
	[.006]	[.006]	[.006]	[.006]	[.007]	[.006]	[.005]	[.008]
White	-.042***	-.042***	-.040***	-.042***	-.040***	-.042***	-.031***	-.037***
	[.008]	[.008]	[.008]	[.008]	[.009]	[.008]	[.007]	[.010]
Black	.021*	.014	.024*	.020	.019	.022*	.015	.027
	[.012]	[.012]	[.013]	[.013]	[.015]	[.012]	[.010]	[.018]
Full time	.075***	.072***	.070***	.075***	.069***	.080***	.052***	.056***
	[.007]	[.008]	[.008]	[.008]	[.009]	[.008]	[.007]	[.012]
Has Children	.024***	.024***	.027***	.024***	.022***	.026***	.019***	.014
	[.006]	[.006]	[.007]	[.007]	[.008]	[.007]	[.006]	[.009]
Is Married	.025***	.026***	.033***	.030***	.020***	.030***	.026***	.034***
	[.006]	[.006]	[.007]	[.007]	[.007]	[.007]	[.006]	[.009]
R-squared	.06	.07	.07	.06	.06	.06	.14	.13
Sample Size	71116	52733	43666	50814	34020	46478	69181	19699
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of an indicator for whether the respondent uses AI daily or frequently during the week (zero otherwise) on a vector of workplace perceptions and whether the firm has a clear AI strategy, and individual covariates and year fixed effects. Workplace (management) practices are based on z-scores of responses to the degree of agreement with the following statements on a 1-5 scale: (a) I feel connected to my organization's culture, (b) My organization consistently delivers on the promises it makes to customers, (c) I trust the leadership of this organization, (d) My organization cares about my overall wellbeing, (e) I have received meaningful feedback in the past week, and (f) I am treated with respect at work. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Table 2: Predicting Having a Clear AI Strategy Using Workplace Practices

	Dep. var. = Has Clear AI Strategy (1/0)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
C: Well-being	.050***						.000
	[.003]						[.008]
C: Connected		.051***					.017**
		[.003]					[.007]
C: Feedback			.060***				.040***
			[.003]				[.006]
C: Customers				.040***			-.000
				[.003]			[.006]
C: Trust					.043***		.016**
					[.004]		[.008]
C: Respect						.042***	-.000
						[.003]	[.006]
Age	-.002***	-.002***	-.002***	-.002***	-.002***	-.002***	-.002***
	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]
Some College	.025***	.029***	.037***	.028***	.025**	.033***	.030**
	[.009]	[.010]	[.010]	[.010]	[.012]	[.010]	[.013]
College or More	.102***	.102***	.110***	.109***	.100***	.102***	.103***
	[.008]	[.009]	[.009]	[.009]	[.011]	[.009]	[.012]
Male	.007	.015**	.009	.010	.015*	.009	.008
	[.006]	[.007]	[.007]	[.007]	[.008]	[.007]	[.009]
White	-.036***	-.036***	-.034***	-.037***	-.030***	-.036***	-.032***
	[.008]	[.009]	[.009]	[.009]	[.010]	[.008]	[.012]
Black	.033***	.023*	.026*	.028**	.044***	.027**	.036*
	[.013]	[.014]	[.014]	[.014]	[.017]	[.013]	[.018]
Full time	.086***	.077***	.065***	.079***	.067***	.077***	.051***
	[.009]	[.010]	[.010]	[.010]	[.012]	[.010]	[.015]
Has Children	.016**	.016**	.016**	.019**	.014	.018**	.017*
	[.007]	[.007]	[.007]	[.008]	[.009]	[.007]	[.010]
Is Married	.006	.006	.009	.007	-.000	.007	-.010
	[.007]	[.007]	[.007]	[.007]	[.008]	[.007]	[.010]
R-squared	.04	.04	.05	.04	.04	.04	.05
Sample Size	59671	41477	39796	39983	26591	42394	19699
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of an indicator for whether the respondent reports that the firm has communicated a clear AI strategy on a vector of workplace perceptions, individual covariates, and time fixed effects. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Workplace (management) practices are based on z-scores of responses to the degree of agreement with the following statements on a 1–5 scale: (a) I feel connected to my organization's culture, (b) My organization consistently delivers on the promises it makes to customers, (c) I trust the leadership of this organization, (d) My organization cares about my overall wellbeing, (e) I have received meaningful feedback in the past week, and (f) I am treated with respect at work. Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Table 3: Effects of AI Utilization and Clear AI Strategy on Employee Outcomes

	Employee Engagement (Q4)				Job Burnout				Job Satisfaction			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Has Clear AI Strategy	.173***	.046**	.016	.051	-.094***	-.003	-.020	.099	.182***	.050**	.035	.012
	[.028]	[.021]	[.021]	[.088]	[.029]	[.025]	[.027]	[.104]	[.028]	[.024]	[.026]	[.123]
Frequent AI Use	-.014	.011	.026	-.004	.115***	.047*	.027	.272**	-.054**	-.036	-.027	-.235**
	[.024]	[.022]	[.024]	[.097]	[.024]	[.026]	[.027]	[.126]	[.024]	[.024]	[.027]	[.094]
× Clear AI Strategy	.213***	.064**	.071**	.153	-.074*	-.048	-.029	-.234	.144***	.041	.039	.220
	[.038]	[.029]	[.030]	[.131]	[.039]	[.034]	[.035]	[.150]	[.038]	[.032]	[.034]	[.151]
Occasional AI Use	.045**	-.013	.015	.013	.030	.009	.009	.091	.021	-.037*	-.028	-.125
	[.020]	[.018]	[.019]	[.066]	[.021]	[.019]	[.020]	[.087]	[.020]	[.020]	[.022]	[.082]
× Clear AI Strategy	.068*	.051*	.083***	-.005	-.039	-.046	-.050	-.002	.037	-.001	.017	.014
	[.037]	[.029]	[.030]	[.122]	[.039]	[.033]	[.034]	[.150]	[.037]	[.032]	[.034]	[.150]
Always Remote	.030	.027	.018	.119	-.007	.026	-.019	-.051	.060**	.098**	.073	.242
	[.024]	[.044]	[.049]	[.166]	[.026]	[.059]	[.063]	[.290]	[.025]	[.050]	[.057]	[.223]
Sometimes Remote	.074***	.066***	.089***	.047	-.040**	-.011	-.036	.171	.119***	.130***	.153***	.195
	[.018]	[.025]	[.027]	[.086]	[.018]	[.036]	[.036]	[.146]	[.017]	[.030]	[.033]	[.125]
R-squared	.03	.82	.83	.76	.03	.80	.81	.71	.03	.77	.78	.67
Sample Size	57649	47874	38831	3782	53757	44409	35996	3538	57440	47664	38664	3762
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Person FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Time FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
SOC3 x Time FE	No	No	Yes	No	No	No	Yes	No	No	No	Yes	No
Sample	Pooled	Pooled	Pooled	Switchers	Pooled	Pooled	Pooled	Switchers	Pooled	Pooled	Pooled	Switchers

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of a z-score for average Q12 employee engagement, a z-score of burnout, and a z-score of job satisfaction on an indicator for whether the organization has articulated a clear AI strategy, indicators of AI utilization frequency, their interactions, and individual covariates and fixed effects. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Frequently Use AI is based on using AI daily or a few times a week; Occasionally Use AI is based on using AI a few times a month or a few times a year; the normalization is relative to those who use it once a year, less than once a year, or never. Employee engagement is defined as a z-score of the average across four of the Q12 indicators due to missing values in the broader set: (Q1) I know what is expected of me at work, (Q3) At work, I have the opportunity to do what I do best every day, (Q6) There is someone at work who encourages my development, (Q7) At work, my opinions seem to count. Burnout is a z-score of: "Please indicate how often each of the following statements is true of your primary job. I feel burned out at work: (a) Always, (b) Very often, (c) Sometimes, (d) Rarely, (e) Never," coded from 5 (always) to 1 (never). Job satisfaction is a z-score of: "How satisfied are you with your place of employment as a place to work?" The switcher sample restricts to respondents with an inferred employer change based on a tenure reset, examining observations within plus or minus two waves of the first observed switch. Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Table 4: Effects of AI Utilization and Clear AI Strategy on Q12 Employee Engagement

Dep. var. =	Expectations	Materials	Do Best	Recognition	Manager	Development	Opinions	Mission	Quality	Best Friend	Progress	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Has Clear AI Strategy	.021	-.012	.051**	.072**	.039	.021	.050**	.093***	.038	.007	.019	.039
	[.025]	[.032]	[.023]	[.031]	[.029]	[.024]	[.022]	[.032]	[.030]	[.029]	[.033]	[.032]
Frequently Use AI	-.039	-.093***	-.002	-.025	-.002	.023	.033	.007	-.012	-.004	-.002	-.011
	[.025]	[.032]	[.024]	[.036]	[.032]	[.024]	[.024]	[.029]	[.033]	[.029]	[.032]	[.030]
× Clear AI Strategy	.027	.150***	.036	.020	.061	.081**	.049	.002	.054	.043	.081*	.060
	[.034]	[.043]	[.033]	[.045]	[.040]	[.032]	[.031]	[.041]	[.043]	[.039]	[.044]	[.041]
Occasionally Use AI	-.026	-.031	-.044**	-.030	-.006	.018	-.001	.003	-.036	-.023	-.005	-.014
	[.022]	[.026]	[.020]	[.027]	[.024]	[.019]	[.019]	[.026]	[.024]	[.024]	[.026]	[.024]
× Clear AI Strategy	.008	.045	.070**	.002	-.009	.046	.032	-.013	.065	.080**	.058	.044
	[.037]	[.041]	[.033]	[.042]	[.041]	[.032]	[.029]	[.042]	[.041]	[.038]	[.043]	[.039]
R-squared	.74	.78	.76	.72	.78	.77	.78	.80	.78	.83	.76	.78
Sample Size	47874	30789	47874	29533	30534	47874	47874	30495	30560	28582	29550	30151
Person FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of a z-score of 12 employee engagement scores on an indicator for whether the firm has a clear AI strategy, indicators of AI utilization, their interactions, and conditional on individual covariates and fixed effects. The Q12 include: (1) I know what is expected of me at work, (2) I have the materials and equipment I need to do my work right, (3) At work, I have the opportunity to do what I do best every day, (4) In the last seven days, I have received recognition or praise for doing good work, (5) My supervisor, or someone at work, seems to care about me as a person, (6) There is someone at work who encourages my development, (7) At work, my opinions seem to count, (8) The mission or purpose of my organization makes me feel my job is important, (9) My associates or fellow employees are committed to doing quality work, (10) I have a best friend at work, (11) In the last six months, someone at work has talked to me about my progress, (12) This last year, I have had opportunities at work to learn and grow. AI Use - Often is based on using AI daily or a few times a week; AI Use - Sometimes is based on using AI a few times a month or a few times a year; the normalization is relative to those who use it once a year, less than once a year, or never. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Table 5: Heterogeneous Effects of AI Utilization and Clear AI Strategy on Employee Engagement

Dep. var. =	Employee Engagement (Q4, z-score)					
	(1)	(2)	(3)	(4)	(5)	(6)
Has Clear AI Strategy	.015	.079**	.041**	.040	.058**	.016
	[.027]	[.031]	[.019]	[.034]	[.026]	[.033]
Frequently Use AI	.016	.002	.012	-.011	.008	.009
	[.028]	[.035]	[.020]	[.040]	[.025]	[.041]
× Clear AI Strategy	.079**	.049	.061**	.085	.027	.134**
	[.038]	[.043]	[.026]	[.057]	[.034]	[.053]
Occasionally Use AI	.017	-.057**	-.005	-.019	-.028	.015
	[.024]	[.027]	[.016]	[.034]	[.021]	[.033]
× Clear AI Strategy	.053	.055	.029	.077	.065*	.021
	[.039]	[.043]	[.025]	[.059]	[.036]	[.048]
R-squared	.83	.80	.81	.83	.82	.81
Sample Size	25972	21840	33410	14093	33705	14085
Person FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Male	Female	>College	<College	White	Non-white

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of a z-score for the average Q12 employee engagement score on an indicator for whether the firm has a clear AI strategy, indicators of AI utilization, their interactions, and conditional on individual covariates and fixed effects, separately by group. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Frequently Use AI is based on using AI daily or a few times a week; Occasionally Use AI is based on using AI a few times a month or a few times a year; the normalization is relative to those who use it once a year, less than once a year, or never. Employee engagement is defined as a z-score of the average across four of the Q12 indicators due to missing values in the broader set: (Q1) I know what is expected of me at work, (Q3) At work, I have the opportunity to do what I do best every day, (Q6) There is someone at work who encourages my development, (Q7) At work, my opinions seem to count. Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Table 6: Effects of AI Utilization and Workplace Trust on Employee Outcomes

	Employee Engagement (Q4)				Job Burnout				Job Satisfaction			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Workplace Trust	.658***	.419***	.415***	.507***	-.406***	-.281***	-.293***	-.414***	.649***	.480***	.471***	.591***
	[.010]	[.019]	[.017]	[.052]	[.013]	[.022]	[.020]	[.066]	[.010]	[.020]	[.020]	[.065]
AI Use - Often	.038*	.045	.032	.118	.104***	.050	.015	.164	-.046**	.009	.006	-.166
	[.020]	[.033]	[.031]	[.107]	[.025]	[.039]	[.037]	[.144]	[.019]	[.029]	[.031]	[.124]
× Workplace Trust	-.026	.033	.054*	-.089	.018	-.054	-.044	-.129	.039*	.024	.049*	-.149
	[.022]	[.028]	[.029]	[.086]	[.027]	[.034]	[.033]	[.133]	[.021]	[.028]	[.029]	[.123]
AI Use - Sometimes	-.006	-.031	-.018	-.141	.038*	-.012	-.006	.105	-.037**	-.009	-.017	-.147
	[.017]	[.024]	[.025]	[.101]	[.021]	[.028]	[.029]	[.126]	[.017]	[.026]	[.028]	[.113]
× Workplace Trust	-.056***	.024	.018	.077	.036*	-.014	.002	.087	-.015	-.003	-.008	-.064
	[.019]	[.025]	[.024]	[.074]	[.022]	[.025]	[.027]	[.098]	[.019]	[.026]	[.028]	[.117]
R-squared	.42	.86	.87	.83	.18	.82	.84	.77	.44	.82	.83	.78
Sample Size	28909	15288	12187	873	27002	13170	10492	765	28823	15218	12134	868
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Person FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Time FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
SOC3 x Time FE	No	No	Yes	No	No	No	Yes	No	No	No	Yes	No
Sample	Pooled	Pooled	Pooled	Switchers	Pooled	Pooled	Pooled	Switchers	Pooled	Pooled	Pooled	Switchers

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of a z-score for average Q12 employee engagement scores, a z-score of burnout, and a z-score of job satisfaction on a z-score of trust in senior leadership, indicators of AI utilization, their interactions, and conditional on individual covariates and fixed effects. AI Use - Often is based on using AI daily or a few times a week; AI Use - Sometimes is based on using AI a few times a month or a few times a year; the normalization is relative to those who use it once a year, less than once a year, or never. Employee engagement is defined as a z-score of the average across the Q12 indicators: (1) I know what is expected of me at work, (2) I have the materials and equipment I need to do my work right, (3) At work, I have the opportunity to do what I do best every day, (4) In the last seven days, I have received recognition or praise for doing good work, (5) My supervisor, or someone at work, seems to care about me as a person, (6) There is someone at work who encourages my development, (7) At work, my opinions seem to count, (8) The mission or purpose of my organization makes me feel my job is important, (9) My associates or fellow employees are committed to doing quality work, (10) I have a best friend at work, (11) In the last six months, someone at work has talked to me about my progress, (12) This last year, I have had opportunities at work to learn and grow. Burnout is a z-score to: "Please indicate how often each of the following statements is true of your primary job (i.e., the job where you spend the most time working). I feel burned out at work: (a) Always, (b) Very often, (c) Sometimes, (d) Rarely, (e) Never" from (5=always) to (1=never). Job satisfaction is similarly defined: "How satisfied are you with your place of employment as a place to work?" Trust in leadership is defined on a 1-5 scale in response to: "I trust the leadership of this organization." Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full and part time workers aged 20–65, and the observations are weighted.

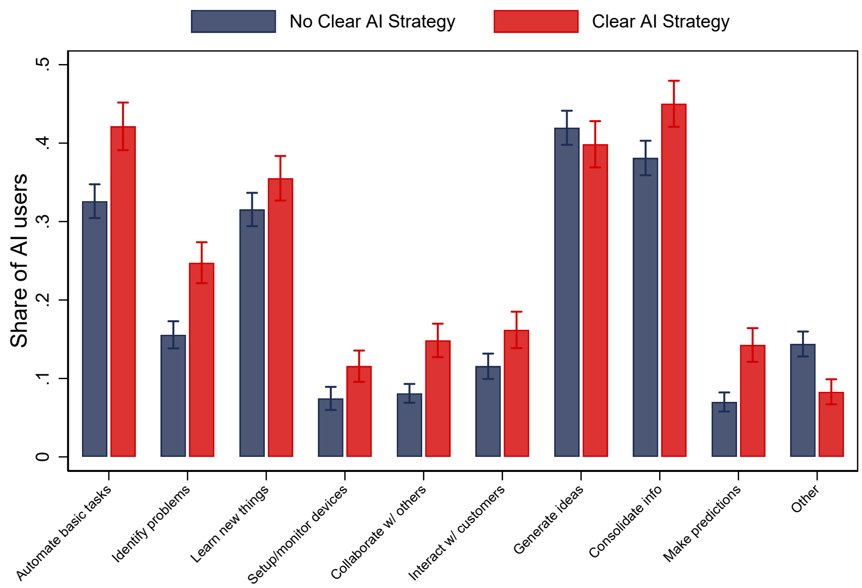
Table 7: Effects of AI Utilization and Workplace Trust on Perceived Productivity from AI

Dep. var. =	Perceived AI Productivity (z-score)			Perceived AI Productivity (1/0)		
	(1)	(2)	(3)	(4)	(5)	(6)
Workplace Trust	.160	-.383	-.441***	-.021	-.336***	-.359***
	[.199]	[.234]	[.169]	[.021]	[.119]	[.094]
AI Use - Often	1.111***	.183	.388	.230***	-.038	-.129
	[.233]	[.396]	[.421]	[.036]	[.193]	[.150]
× Workplace Trust	.037	.610**	.718***	.095***	.392***	.408***
	[.202]	[.270]	[.251]	[.029]	[.128]	[.127]
AI Use - Sometimes	.524**	-.008	-.027	.029	-.078	-.200
	[.234]	[.396]	[.355]	[.032]	[.182]	[.131]
× Workplace Trust	.027	.438	.246	.044*	.293**	.337***
	[.202]	[.365]	[.203]	[.024]	[.124]	[.098]
R-squared	.17	.73	.88	.12	.76	.86
Sample Size	2856	344	266	2856	344	266
Demographics	Yes	Yes	Yes	Yes	Yes	Yes
Person FE	No	Yes	Yes	No	Yes	Yes
Time FE	No	Yes	Yes	No	Yes	Yes
SOC3 x Time FE	No	No	Yes	No	No	Yes

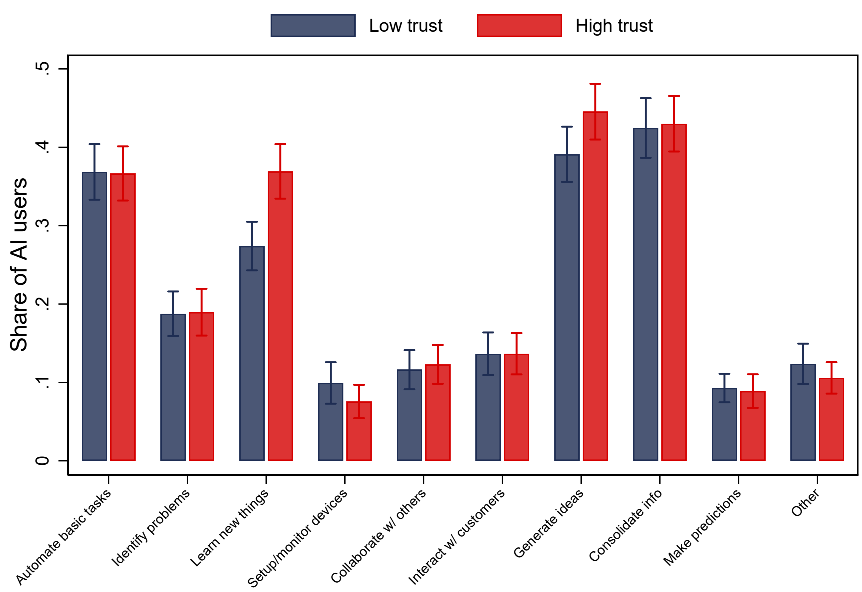
Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of perceived productivity on AI on a z-score of trust in the workplace, indicators of AI utilization, their interactions, and conditional on individual covariates and fixed effects. AI Use - Often is based on using AI daily or a few times a week; AI Use - Sometimes is based on using AI a few times a month or a few times a year; the normalization is relative to those who use it once a year, less than once a year, or never. Perceived AI productivity is measured as a z-score of a 1-5 index, as well as an indicator based on a 5/5, in response to the following question: "What type of impact do you think these artificial intelligence (AI) tools will have on your individual productivity and efficiency at work?" The responses on perceived AI productivity range from extremely negative to extremely positive. Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full and part time workers aged 20–65, and the observations are weighted.

Figure 7: Composition of AI Use-cases by Employees, 2024-2025

Panel A: By Clear AI Strategy



Panel B: Trust in Leadership



Notes.—Sources: Gallup Panel, 2024-2025. The figure plots the share of respondents across different AI applications distinguishing among respondents who report having a clear AI strategy at their firm (Panel A), and high versus low trust in leadership, i.e. a score of 4/5 or 5/5 (Panel B). Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Trust in leadership is defined on a 1-5 scale in response to: "I trust the leadership of this organization." The sample is restricted to full and part time workers aged 20–65, those who use AI at least somewhat, and the observations are weighted.

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A Online Appendix

A.1 Extensions of the Stylized Model

A.1.1 Closed-form Example, Corner Solutions, and Comparative Statics

Consider the commonly used quadratic cost specification

$$c(a;\theta) = \frac{a^2}{2\theta}, \qquad \theta > 0, \tag{8}$$

and let the per-task benefit be linear in embedding with a complementarity between embedding and managerial clarity:

$$B(E,S) = \alpha + \beta E + \gamma S E, \qquad \alpha, \beta, \gamma \geq 0, \tag{9}$$

where $S \in \{0,1\}$ (or $S \in [0,1]$ in a continuous interpretation).

Interior solution. For $a \in (0,1)$, the first-order condition (4) becomes

$$B(E,S) = \frac{a^*}{\theta}, \tag{10}$$

so the interior optimum is

$$a^*(E,S,\theta) = \theta\, B(E,S). \tag{11}$$

Corner solutions. Because $a \in [0,1]$, the equilibrium is

$$a^*(E,S,\theta) = \min\{1,\ \theta B(E,S)\}. \tag{12}$$

Thus, full adoption occurs when $\theta B(E,S) \geq 1$.

Comparative statics in closed form. On the interior region $0 < \theta B(E,S) < 1$, we have

$$\frac{\partial a^*}{\partial \theta} = B(E,S) > 0, \tag{13}$$
$$\frac{\partial a^*}{\partial E} = \theta(\beta + \gamma S) > 0, \tag{14}$$
$$\frac{\partial a^*}{\partial S} = \theta\gamma E \geq 0, \tag{15}$$
$$\frac{\partial^2 a^*}{\partial E \partial S} = \theta\gamma > 0 \quad \text{(complementarity)}. \tag{16}$$

Closed-form outcome examples. The main text imposes a complementarity structure for outcomes. Two simple specifications show how these cross-partials arise naturally.

First, let per-worker productivity be

$$Y = Y_0 + a \cdot B(E,S). \tag{17}$$

Under (11), this becomes $Y = Y_0 + \theta[B(E,S)]^2$ on the interior region, so the realized productivity gains from embedding and clarity scale with psychological safety through equilibrium adoption.

Second, let an employee outcome (e.g., engagement) be

$$G = G_0 + \eta\, a\, \theta + \zeta\, a\, S, \qquad \eta, \zeta > 0, \tag{18}$$

which implies $\partial^2 G/\partial a\,\partial\theta = \eta > 0$ and $\partial^2 G/\partial a\,\partial S = \zeta > 0$ as in (5). This toy specification is useful for interpretation (e.g., adoption is more beneficial when experimentation is safe and guidance is clear) without restricting the empirical analysis to a particular functional form.

Proof of Lemma 1. Let

$$U(a; E, S, \theta) \equiv a\,B(E, S) - c(a; \theta), \qquad a \in [0, 1].$$

Because $c_{aa}(a;\theta) > 0$, the objective U is strictly concave in a, so the worker's problem (3) has a unique optimal choice $a^*(E, S, \theta)$.

To establish global monotonicity, note that U has increasing differences in (a, E), (a, S), and (a, θ):

$$U_{aE} = B_E(E, S) > 0, \qquad U_{aS} = B_S(E, S) \geq 0, \qquad U_{a\theta} = -c_{a\theta}(a; \theta) > 0.$$

Hence, by monotone comparative statics, the unique optimal adoption choice $a^*(E, S, \theta)$ is weakly increasing in E, S, and θ.

Now consider any interior region where $a^*(E, S, \theta) \in (0, 1)$. The first-order condition (4) is

$$F(a, E, S, \theta) \equiv c_a(a; \theta) - B(E, S) = 0.$$

Strict convexity implies $c_{aa}(a;\theta) > 0$, so $F_a = c_{aa} > 0$. By the Implicit Function Theorem, a^* is continuously differentiable in (E, S, θ) locally and

$$\frac{\partial a^*}{\partial E} = -\frac{F_E}{F_a} = \frac{B_E(E,S)}{c_{aa}(a^*;\theta)} > 0, \tag{19}$$

$$\frac{\partial a^*}{\partial S} = -\frac{F_S}{F_a} = \frac{B_S(E,S)}{c_{aa}(a^*;\theta)} \geq 0, \tag{20}$$

$$\frac{\partial a^*}{\partial \theta} = -\frac{F_\theta}{F_a} = -\frac{c_{a\theta}(a^*;\theta)}{c_{aa}(a^*;\theta)} > 0, \tag{21}$$

where the last inequality uses $c_{a\theta} < 0$ (higher psychological safety lowers the marginal cost of adoption). For the complementarity between embedding and clarity, differentiate (19) with respect to S:

$$\frac{\partial^2 a^*}{\partial S \partial E} = \frac{B_{ES}(E,S)}{c_{aa}(a^*;\theta)} - \frac{B_E(E,S)\, c_{aaa}(a^*;\theta)}{[c_{aa}(a^*;\theta)]^2} \cdot \frac{\partial a^*}{\partial S}. \tag{22}$$

A sufficient (and standard) condition for $\partial^2 a^*/\partial S \partial E > 0$ on the interior is $B_{ES} > 0$ together with either (i) c_{aaa} small relative to c_{aa} in the relevant region, or (ii) $c_{aaa} \leq 0$ (weakly decreasing curvature), or (iii) the quadratic benchmark (8) where $c_{aaa} \equiv 0$, yielding

$$\frac{\partial^2 a^*}{\partial S \partial E} = \frac{B_{ES}(E,S)}{c_{aa}(a^*;\theta)} > 0.$$

Thus, $a^*(E, S, \theta)$ is weakly increasing in E, S, and θ globally, while the derivative formulas above hold on interior regions where $a^* \in (0, 1)$. □

A.1.2 Proof of Proposition 1

Proof of Proposition 1. Let $W = W(a, E, S, \theta)$ be twice continuously differentiable in (a, S, θ) for fixed E, and suppose (5) holds. Fix E. For any given adoption level a and communication environment S, define the marginal effect of adoption on outcomes as

$$M(a,\theta;E,S) \equiv \frac{\partial W(a,E,S,\theta)}{\partial a}.$$

Then, holding a, E, and S fixed,

$$\frac{\partial M(a,\theta;E,S)}{\partial \theta} = \frac{\partial^2 W(a,E,S,\theta)}{\partial a\,\partial \theta} > 0.$$

Thus, the marginal effect of AI adoption on outcomes is increasing in θ. Similarly, holding a, E, and θ fixed,

$$\frac{\partial}{\partial S}\left(\frac{\partial W(a,E,S,\theta)}{\partial a}\right) = \frac{\partial^2 W(a,E,S,\theta)}{\partial a\,\partial S} > 0,$$

so the marginal effect of AI adoption on outcomes is also increasing in S. Hence, for any given level of adoption a, the marginal effect of AI usage is larger in high-θ and high-S environments. □

A.1.3 Microfoundation for S as Communication Precision

This subsection provides a reduced-form microfoundation for interpreting S as the precision of internal communication about AI use-cases, constraints, and standards, consistent with the view that communication inside firms is a design choice and is shaped by organizational frictions (Dessein, 2002; Alonso et al., 2008) and by language/communication costs (Crémer et al., 2007).

Let the productivity gain from using AI depend on whether usage is aligned with a firm-specific "target" practice τ (e.g., approved tools, data-handling rules, quality standards). Suppose workers choose a and also (implicitly) a practice x; misalignment yields a coordination loss. Let x be a worker's belief/action about "how to use AI." The manager sends a message m about τ; workers observe a noisy signal

$$x = \tau + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2(S)),$$

where higher clarity S reduces noise, so $(\sigma^2)'(S) < 0$. Let $\kappa(E) > 0$ denote the importance of alignment for realized returns. If the expected per-task gross benefit is $\bar{b}(E)$ but misalignment reduces effective benefit by $\kappa(E)E[(x-\tau)^2] = \kappa(E)\sigma^2(S)$, then the net per-task return becomes

$$B(E, S) = \bar{b}(E) - \kappa(E)\sigma^2(S). \tag{23}$$

If embedding increases the underlying return $\bar{b}(E)$ and also raises the value of alignment—for example, because AI is more deeply embedded in shared workflows, so deviations from the target practice generate larger downstream coordination losses—then $\kappa = \kappa(E)$ with $\kappa'(E) > 0$. In that case,

$$B_{ES}(E, S) = -\kappa'(E)\,(\sigma^2)'(S) > 0,$$

since $\kappa'(E) > 0$ and $(\sigma^2)'(S) < 0$. Moreover,

$$B_E(E, S) = \bar{b}'(E) - \kappa'(E)\sigma^2(S),$$

so the maintained assumption $B_E > 0$ holds provided the direct productivity effect of embedding dominates the increase in alignment stakes. Thus, clearer communication and embedding are complements in returns, as assumed in the main text.

A.1.4 Task Interdependence and Clarity (H3)

To connect more to H3, let $\tau \geq 0$ index task interdependence/coordination intensity at the occupation (or team) level. Suppose the coordination loss from unclear guidance scales with τ:

$$B(E, S; \tau) = \bar{b}(E) - \tau\, \kappa(E)\, \sigma^2(S), \tag{24}$$

with $\kappa(E) > 0$, and where higher clarity reduces noise so $(\sigma^2)'(S) < 0$. Then

$$\frac{\partial B}{\partial S} = -\tau\, \kappa(E)\, (\sigma^2)'(S) \geq 0, \qquad \frac{\partial^2 B}{\partial S\, \partial \tau} = -\kappa(E)\, (\sigma^2)'(S) > 0,$$

with $\partial B/\partial S > 0$ whenever $\tau > 0$. Thus, the marginal return to clarity S is larger in high-interdependence jobs. Under the quadratic cost benchmark (8) together with (24), this implies larger $\partial a^*/\partial S$ and larger outcome interactions (e.g., larger AI $\times$ ClearAI) at higher τ, which motivates the triple-interaction test in H3.

A.1.5 Credibility and the Triple Complementarity (H4)

To formalize the idea that trust increases the *effectiveness* of communication (beyond shifting experimentation costs), let the manager's clarity choice be "discounted" when trust is low. A parsimonious way is to let effective clarity be $S \cdot g(\theta)$, where g is increasing and concave, $g'(\theta) > 0$, $g(0) = 0$, and $g(1) = 1$. Then

$$B(E, S, \theta) = \alpha + \beta E + \gamma E \cdot S \cdot g(\theta). \tag{25}$$

Under (8), equilibrium adoption is $a^* = \theta B(E, S, \theta)$ (interior). Differentiating yields

$$\frac{\partial^2 a^*}{\partial S\,\partial\theta} = \gamma E\left[g(\theta) + \theta g'(\theta)\right] > 0,$$

and outcome complementarities become naturally triple-interaction objects. This extension provides a structural rationale for why the AI $\times$ ClearAI interaction should be strongest when trust is high (H4).

A.1.6 Role Clarity via Ambiguity Costs (H6)

To generate the sign-switching prediction in H6 within the model, let worker well-being/role clarity include an ambiguity term that increases with adoption when guidance is unclear. For example, let an outcome R (role clarity) be

$$R = R_0 + \underbrace{\rho_1 a S}_{\text{guided adoption clarifies}} - \underbrace{\rho_0 a(1-S)}_{\text{unguided adoption confuses}}, \qquad \rho_0, \rho_1 > 0. \tag{26}$$

Then

$$\frac{\partial R}{\partial a} = \begin{cases} +\rho_1 & \text{if } S = 1, \\ -\rho_0 & \text{if } S = 0, \end{cases}$$

which produces the crossover interaction: adoption increases role clarity under high clarity and decreases it under low clarity.

A.1.7 Composition of AI Utilization (H5)

To rationalize H5, consider two AI application types: a standardized/workflow-embedded use a_s and an exploratory/learning use a_e, with $a_s + a_e \leq 1$. Suppose the worker solves

$$\max_{a_s, a_e \geq 0,\, a_s + a_e \leq 1} \left\{ a_s B_s(E, S) + a_e B_e(E) - \left[\frac{a_s^2}{2\theta_s(\theta)} + \frac{a_e^2}{2\theta_e(\theta)} \right] \right\},$$

where standardized uses are more alignment-intensive, so

$$\frac{\partial B_s(E, S)}{\partial S} > 0,$$

while exploratory uses are much less sensitive to clarity, so $B_e(E)$ does not depend directly on S. Assume a local interior region with

$$a_s^* > 0, \qquad a_e^* > 0, \qquad a_s^* + a_e^* < 1.$$

Then the first-order conditions imply

$$a_s^* = \theta_s(\theta) B_s(E, S), \qquad a_e^* = \theta_e(\theta) B_e(E).$$

Hence,

$$\frac{a_e^*}{a_s^*} = \frac{\theta_e(\theta)}{\theta_s(\theta)} \cdot \frac{B_e(E)}{B_s(E, S)}.$$

For fixed E and θ, higher S lowers a_e^*/a_s^* whenever $\partial B_s/\partial S > 0$, so clarity tilts usage toward standardized applications. For fixed E and S, higher θ tilts usage toward exploratory applications provided

$$\frac{d}{d\theta}\left(\frac{\theta_e(\theta)}{\theta_s(\theta)}\right) > 0,$$

equivalently,

$$\theta_s(\theta)\theta_e'(\theta) - \theta_e(\theta)\theta_s'(\theta) > 0.$$

This condition says that psychological safety relaxes the effective cost of exploratory use proportionally more than that of standardized use. These comparative statics generate the composition predictions in H5 and map directly into use-case heterogeneity tests.

A.1.8 Endogenous (E, S, θ) and Organizational Complementarities

The main text treats (E, S, θ) as given when characterizing worker adoption. If desired, one can endogenize these choices to formalize organizational complementarities (in the spirit of Milgrom and Roberts, 1990; Bresnahan et al., 2002). Let the firm choose E and the manager choose (S, θ) to maximize

$$\Pi(E, S, \theta) = \underbrace{\Omega(a^*(E, S, \theta), E, S, \theta)}_{\text{gross surplus}} - K_E(E) - K_S(S) - K_\theta(\theta),$$

with convex costs $K_\cdot$. On interior regions where $a^*(E, S, \theta)$ is differentiable, a sufficient set of conditions for complementary choices is:

$$a_E^*,\ a_S^*,\ a_\theta^* \geq 0, \qquad a_{ES}^*,\ a_{E\theta}^*,\ a_{S\theta}^* \geq 0,$$

and

$$\Omega_a,\ \Omega_{aa},\ \Omega_{aE},\ \Omega_{aS},\ \Omega_{a\theta},\ \Omega_{ES},\ \Omega_{E\theta},\ \Omega_{S\theta} \geq 0.$$

Under these conditions,

$$\Pi_{ES} = \Omega_a a^*_{ES} + \Omega_{aa} a^*_E a^*_S + \Omega_{aE} a^*_S + \Omega_{aS} a^*_E + \Omega_{ES} \geq 0,$$

$$\Pi_{E\theta} = \Omega_a a^*_{E\theta} + \Omega_{aa} a^*_E a^*_\theta + \Omega_{aE} a^*_\theta + \Omega_{a\theta} a^*_E + \Omega_{E\theta} \geq 0,$$

$$\Pi_{S\theta} = \Omega_a a^*_{S\theta} + \Omega_{aa} a^*_S a^*_\theta + \Omega_{aS} a^*_\theta + \Omega_{a\theta} a^*_S + \Omega_{S\theta} \geq 0.$$

Because the cost terms are separable, they do not affect these cross-partials. Thus, Π is supermodular in (E, S, θ) under these sufficient conditions, so optimal choices are complementary: higher S raises the marginal return to E, and higher θ raises the marginal return to both E and S (especially under the credibility formulation (25)). This provides an organizational interpretation of observed cross-firm heterogeneity: firms that invest more in embedding are also those that tend to communicate more clearly and cultivate greater psychological safety because these choices are jointly optimal when the technology is valuable.

A.2 Additional Details on Data and Measurement

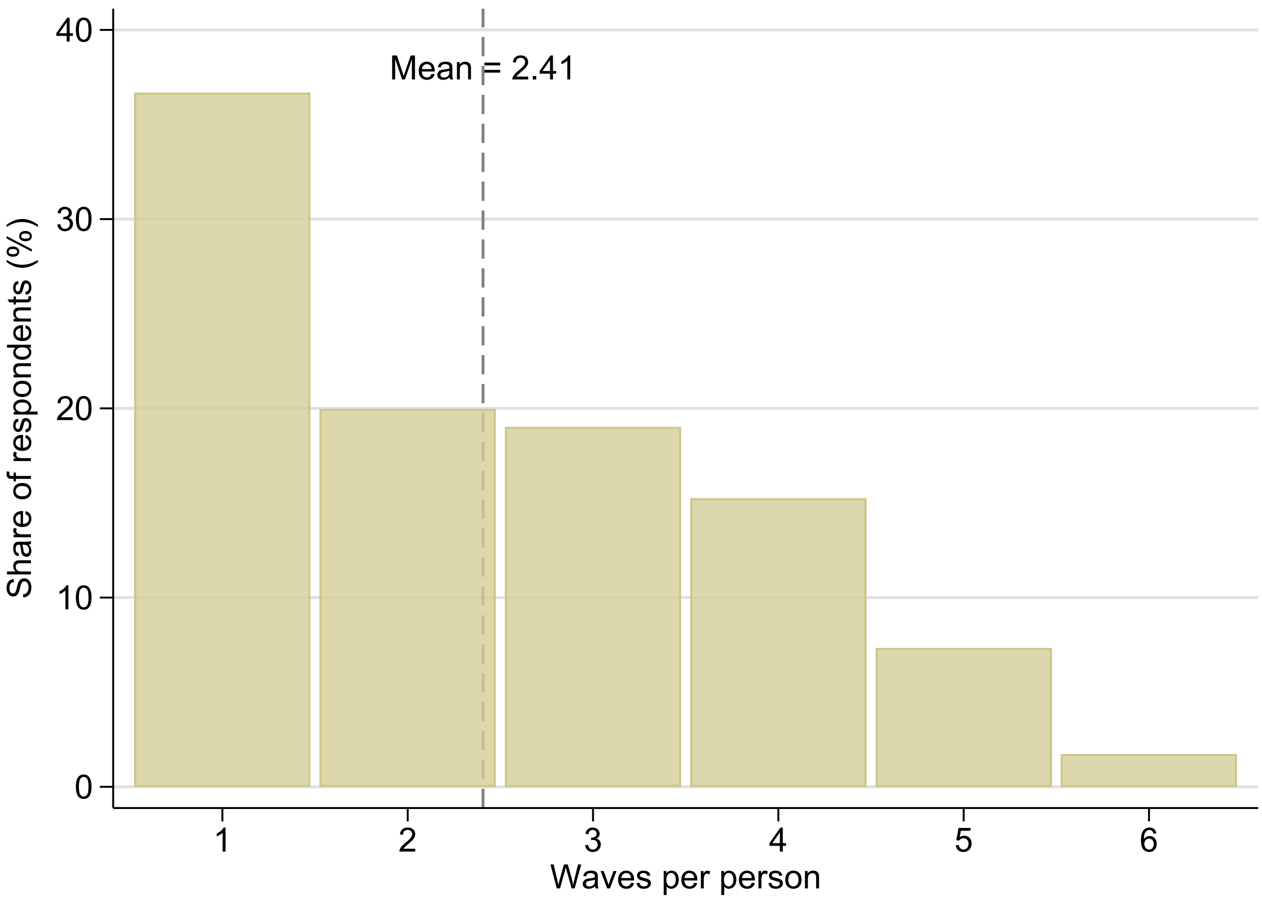


Figure A.1: Descriptive Evidence on Longitudinal Variation

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The figure plots the share of respondents across the potential number of waves in the data.

Table A.1: Panel Structure and AI-Use Transitions

Panel A: Panel Structure			
Unique workers		33,743	
Mean waves per worker		2.34	
Share with 2+ waves (%)		62.6	
Share ever switching AI use category (%)		26.8	
Share ever switching, among 2+ wave workers (%)		42.9	
Adjacent-wave transitions		46,067	
Adoption rate: Never → (Occasional/Frequent) (%)		13.5	
Cessation rate: (Occasional/Frequent) → Never (%)		5.5	
Adoption-to-cessation ratio		2.46	
Panel B: Adjacent-Wave Transition Matrix (% of row)			
	Never	Occasional	Frequent
Never	75.4	16.2	8.4
Occasional	17.5	54.6	27.9
Frequent	7.1	15.5	77.3

Notes.—Source: Gallup Panel, 2023–Q1:2026. Panel A reports person-level statistics for the analytic sample restricted to respondents aged 20–65 with non-missing AI use. A switcher is a worker whose AI use category changes in at least one adjacent wave pair. The adoption rate is the share of adjacent-wave transitions in which a respondent moves from Never to Occasional or Frequent; the cessation rate is the share moving from Occasional or Frequent to Never. Panel B reports the row-conditional distribution of adjacent-wave transitions across three mutually exclusive AI use categories: Never (once a year, less than once a year, or never), Occasional (a few times a month or a few times a year), and Frequent (daily or a few times a week); each row sums to 100. Observations are weighted.

Table A.2: Summary Statistics on the Gallup Workplace Panel

	2023		2024		2025		2026	
	(1)		(2)		(3)		(4)	
	mean	sd	mean	sd	mean	sd	mean	sd
Frequent AI Use	0.10	0.30	0.12	0.32	0.22	0.41	0.26	0.44
Occasional AI Use	0.10	0.29	0.15	0.36	0.19	0.39	0.21	0.41
Never AI Use	0.81	0.39	0.73	0.44	0.52	0.50	0.47	0.50
Very Likely AI Will Displace Job	0.03	0.18	0.03	0.16	0.03	0.18	0.04	0.20
Somewhat Likely AI Will Displace Job	0.14	0.35	0.15	0.35	0.15	0.35	0.18	0.39
Hours Worked/Week	38.3	12.7	37.7	12.2	39.1	11.4	38.8	11.6
Age	43.9	13.8	43.1	14.3	43.6	14.0	43.4	14.0
Male	0.53	0.50	0.53	0.50	0.53	0.50	0.53	0.50
White	0.67	0.47	0.73	0.44	0.65	0.48	0.64	0.48
Black	0.12	0.32	0.10	0.30	0.12	0.32	0.12	0.32
Full-time	0.78	0.41	0.77	0.42	0.82	0.38	0.83	0.37
Part-time	0.22	0.41	0.23	0.42	0.18	0.38	0.17	0.37
Has Children	0.35	0.48	0.33	0.47	0.34	0.48	0.34	0.47
Married	0.60	0.49	0.53	0.50	0.52	0.50	0.52	0.50
Less than HS	0.32	0.47	0.28	0.45	0.31	0.46	0.31	0.46
Some College	0.26	0.44	0.29	0.46	0.26	0.44	0.26	0.44
College or More	0.42	0.49	0.42	0.49	0.43	0.49	0.43	0.49
Observations	9322		9966		52729		22563	

Notes.—Sources: Gallup Panel, 2023–2026. The table reports summary statistics on the key variables by year. AI Use - Often is based on using AI daily or a few times a week; AI Use - Sometimes is based on using AI a few times a month or a few times a year; AI Use - Never is based on using AI once a year, less than once a year, or never. Very Likely AI Will Displace Job is an indicator for believing that AI is "very likely" to replace one's job within the next five years; Somewhat Likely AI Will Displace Job is an indicator for believing displacement is "somewhat likely" or "very likely." Both displacement indicators are coded as missing when the underlying question is missing. Demographics include: age, race (White, Black), full-time status, having children, and being married. The sample is restricted to full- and part-time workers ages 20-65, and the observations are weighted.

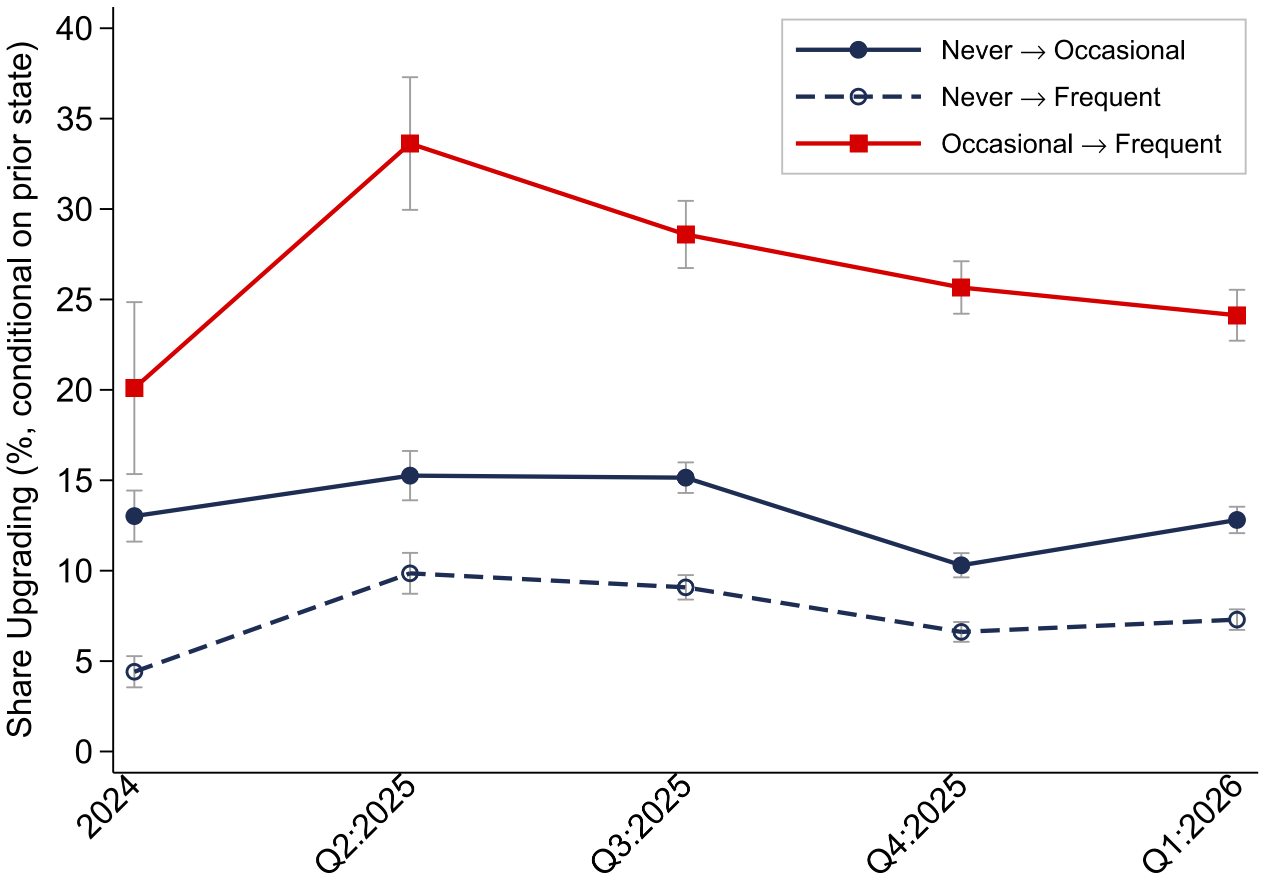


Figure A.2: Within-person Adoption of AI

Notes.—Sources: Gallup Panel, 2023–2026. The figure plots within-person AI adoption transition rates across consecutive survey waves, conditional on the respondent's AI use state in the prior wave. AI use is consolidated into three categories based on responses to "How often do you use artificial intelligence in your role": frequent (daily or a few times a week), occasional (a few times a month or a few times a year), and never (once a year, less often than once a year, or never). Each series shows the share of respondents in a given origin state who upgraded to a higher category by the next wave: Never → Occasional (solid navy), Never → Frequent (dashed navy), and Occasional → Frequent (solid red). The x-axis denotes the arrival wave; the 2023 wave does not appear because it serves as the baseline from which the first transition is measured. Before asking this question, respondents are given a definition of AI: "Artificial intelligence is when a computer does or enhances a task that is normally performed by a person." The sample is restricted to full- and part-time workers aged 20–65 who are observed in at least two waves, and the observations are weighted. Capped spikes denote 95% confidence intervals.

A.3 Robustness to the Main Results

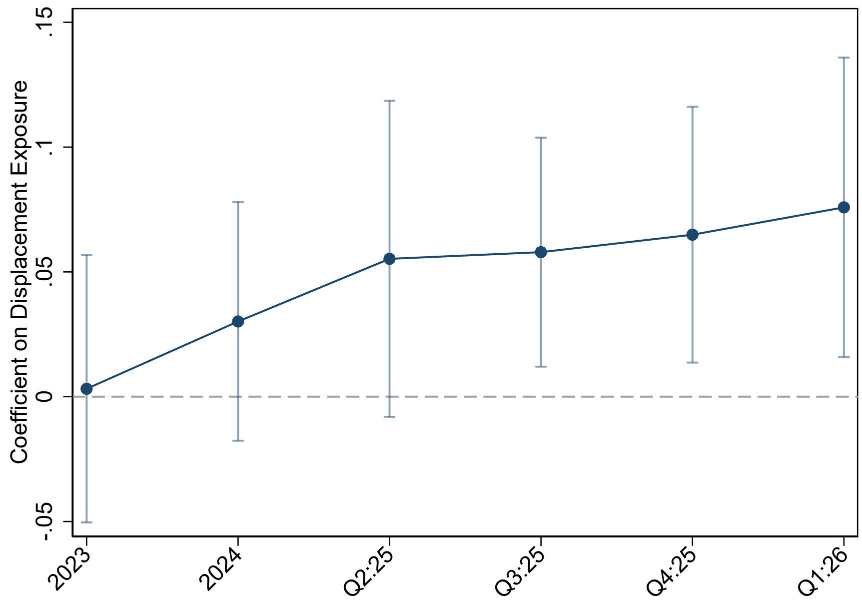
Table A.3: Workplace Determinants of AI Utilization (With Stable Sample)

	Dep. var. = Use AI Frequently (1/0)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
C: Well-being	.022***							-.011
	[.004]							[.007]
C: Connected		.026***						.006
		[.004]						[.006]
C: Feedback			.029***					.008
			[.004]					[.005]
C: Customers				.016***				-.007
				[.004]				[.005]
C: Trust					.028***			.016**
					[.004]			[.007]
C: Respect						.022***		.005
						[.004]		[.006]
Has Clear AI Strategy							.274***	.268***
							[.012]	[.012]
Age	-.003***	-.003***	-.003***	-.003***	-.003***	-.003***	-.002***	-.002***
	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]
Some College	.017	.018	.017	.017	.017	.015	.009	.009
	[.013]	[.013]	[.013]	[.013]	[.013]	[.013]	[.012]	[.012]
College or More	.119***	.119***	.119***	.121***	.120***	.116***	.091***	.090***
	[.012]	[.012]	[.012]	[.012]	[.012]	[.012]	[.012]	[.012]
Male	.015*	.016*	.015*	.014	.015*	.013	.013	.013
	[.009]	[.009]	[.009]	[.009]	[.009]	[.009]	[.008]	[.008]
White	-.046***	-.046***	-.045***	-.047***	-.046***	-.047***	-.037***	-.037***
	[.011]	[.011]	[.011]	[.011]	[.011]	[.011]	[.010]	[.010]
Black	.037**	.037**	.037**	.038**	.038**	.037**	.027	.027
	[.019]	[.019]	[.019]	[.019]	[.019]	[.019]	[.018]	[.018]
Full time	.069***	.069***	.067***	.070***	.072***	.068***	.055***	.056***
	[.013]	[.013]	[.013]	[.013]	[.013]	[.013]	[.012]	[.012]
Has Children	.020**	.019**	.020**	.020**	.019**	.021**	.016*	.014
	[.010]	[.009]	[.009]	[.010]	[.009]	[.009]	[.009]	[.009]
Is Married	.031***	.031***	.033***	.033***	.031***	.030***	.035***	.034***
	[.009]	[.009]	[.009]	[.009]	[.009]	[.009]	[.009]	[.009]
R-squared	.05	.05	.06	.05	.06	.05	.13	.13
Sample Size	19699	19699	19699	19699	19699	19699	19699	19699
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

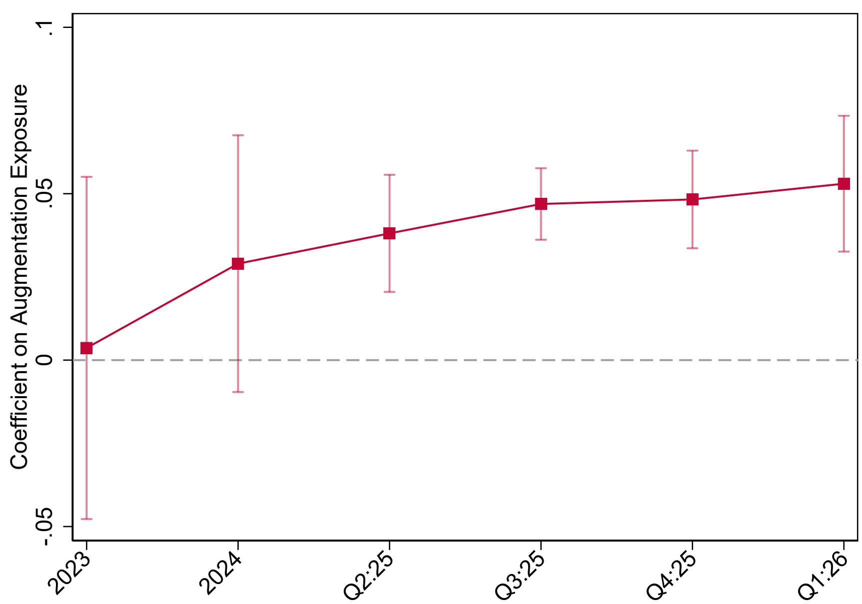
Notes.—Sources: Gallup Panel, 2023–2026. The table reports the coefficients associated with regressions of an indicator for whether the respondent uses AI daily or frequently during the week (zero otherwise) on a vector of workplace perceptions and whether the firm has a clear AI strategy, and individual covariates and year fixed effects. Workplace (management) practices are based on responses to the degree of agreement with the following statements on a 1-5 scale: (a) I feel connected to my organization's culture, (b) My organization consistently delivers on the promises it makes to customers, (c) I trust the leadership of this organization, (d) My organization cares about my overall wellbeing, (e) I have received meaningful feedback in the past week, and (f) I am treated with respect at work. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, as well as to non-missing values on each of the culture variables, and the observations are weighted.

Figure A.3: Exposure to Generative AI and Frequent AI Use Over Time

Panel A: Displacement Exposure



Panel B: Augmentation Exposure



Notes.—Sources: Eloundou et al. (2024) and Gallup Panel, 2023–Q1:2026. The figure plots the coefficients on the interaction between occupation-level generative AI exposure and time period indicators from regressions of an indicator for frequent AI use (daily or a few times a week) on exposure × period interactions, demographic controls, and period fixed effects. Panel A uses the displacement exposure measure (β rating) from Eloundou et al. (2024); Panel B uses the augmentation exposure measure, defined as twice the difference between the β and α ratings. Both exposure measures are standardized to mean zero and unit variance and are matched to respondents at the three-digit SOC level. The base period is 2023; coefficients for subsequent periods represent the change in the exposure–adoption gradient relative to 2023. Capped spikes denote 95% confidence intervals based on standard errors clustered at the three-digit SOC level. Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Table A.4: Predicting Having a Clear AI Strategy Using Workplace Practices (With Stable Sample)

	Dep. var. = Has Clear AI Strategy (1/0)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
C: Well-being	.048***						.000
	[.004]						[.008]
C: Connected		.051***					.017**
		[.004]					[.007]
C: Feedback			.059***				.040***
			[.004]				[.006]
C: Customers				.037***			-.000
				[.004]			[.006]
C: Trust					.049***		.016**
					[.004]		[.008]
C: Respect						.041***	-.000
						[.004]	[.006]
Age	-.002***	-.003***	-.002***	-.003***	-.003***	-.003***	-.002***
	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]
Some College	.029**	.032**	.029**	.030**	.030**	.026*	.030**
	[.013]	[.013]	[.013]	[.013]	[.013]	[.013]	[.013]
College or More	.103***	.103***	.104***	.107***	.105***	.097***	.103***
	[.012]	[.012]	[.012]	[.012]	[.012]	[.012]	[.012]
Male	.006	.008	.006	.005	.006	.003	.008
	[.009]	[.009]	[.009]	[.009]	[.009]	[.009]	[.009]
White	-.034***	-.033***	-.031***	-.035***	-.034***	-.035***	-.032***
	[.012]	[.012]	[.012]	[.012]	[.012]	[.012]	[.012]
Black	.036**	.035*	.035*	.039**	.037**	.036*	.036*
	[.019]	[.019]	[.019]	[.019]	[.018]	[.019]	[.018]
Full time	.053***	.053***	.047***	.054***	.058***	.051***	.051***
	[.015]	[.015]	[.015]	[.015]	[.015]	[.015]	[.015]
Has Children	.019*	.017*	.019*	.019*	.017*	.020**	.017*
	[.010]	[.010]	[.010]	[.010]	[.010]	[.010]	[.010]
Is Married	-.011	-.011	-.007	-.008	-.010	-.012	-.010
	[.010]	[.010]	[.010]	[.010]	[.010]	[.010]	[.010]
R-squared	.04	.05	.05	.04	.04	.04	.05
Sample Size	19699	19699	19699	19699	19699	19699	19699
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of an indicator for whether the respondent reports that the firm has communicated a clear AI strategy on a vector of workplace perceptions, individual covariates, and time fixed effects. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Workplace (management) practices are based on responses to the degree of agreement with the following statements on a 1–5 scale: (a) I feel connected to my organization's culture, (b) My organization consistently delivers on the promises it makes to customers, (c) I trust the leadership of this organization, (d) My organization cares about my overall wellbeing, (e) I have received meaningful feedback in the past week, and (f) I am treated with respect at work. Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, as well as to non-missing values on each of the culture variables, and the observations are weighted.

Table A.5: Workplace Determinants of AI Utilization (With Displacement Risk)

	Dep. var. = Use AI Frequently (1/0)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AI Will Displace Job	.149***	.150***	.146***	.147***	.133***	.146***	.100***	.088***
	[.011]	[.011]	[.011]	[.011]	[.019]	[.011]	[.010]	[.029]
C: Well-being	.028***							-.017
	[.003]							[.014]
C: Connected		.031***						-.001
		[.003]						[.010]
C: Feedback			.045***					.010
			[.004]					[.009]
C: Customers				.018***				-.008
				[.003]				[.011]
C: Trust					.016***			.037**
					[.005]			[.015]
C: Respect						.037***		.000
						[.004]		[.011]
Has Clear AI Strategy							.282***	.267***
							[.009]	[.031]
Age	-.002***	-.002***	-.001***	-.002***	-.002***	-.002***	-.001***	-.002***
	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]	[.000]	[.001]
Some College	.030***	.028***	.052***	.027***	-.010	.050***	.029***	.003
	[.010]	[.010]	[.010]	[.010]	[.015]	[.010]	[.009]	[.024]
College or More	.131***	.128***	.169***	.131***	.045***	.166***	.113***	.024
	[.009]	[.010]	[.010]	[.010]	[.014]	[.009]	[.009]	[.022]
Male	.028***	.029***	.025***	.028***	.021**	.023***	.025***	-.011
	[.006]	[.007]	[.007]	[.007]	[.009]	[.007]	[.006]	[.015]
White	-.038***	-.038***	-.038***	-.037***	-.042**	-.039***	-.029***	-.079***
	[.009]	[.010]	[.009]	[.010]	[.016]	[.009]	[.008]	[.025]
Black	.003	.002	.012	.009	-.008	.013	.010	-.010
	[.014]	[.014]	[.015]	[.015]	[.022]	[.014]	[.013]	[.039]
Full time	.074***	.073***	.076***	.076***	.070***	.085***	.051***	.056***
	[.009]	[.009]	[.010]	[.009]	[.012]	[.009]	[.008]	[.021]
Has Children	.025***	.024***	.026***	.026***	.021**	.025***	.018***	.010
	[.007]	[.007]	[.008]	[.008]	[.011]	[.008]	[.007]	[.017]
Is Married	.025***	.026***	.034***	.027***	.007	.033***	.028***	.028*
	[.007]	[.007]	[.008]	[.007]	[.010]	[.008]	[.007]	[.016]
R-squared	.09	.09	.09	.09	.06	.09	.16	.15
Sample Size	33000	32677	25012	31435	14785	26283	30558	3548
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

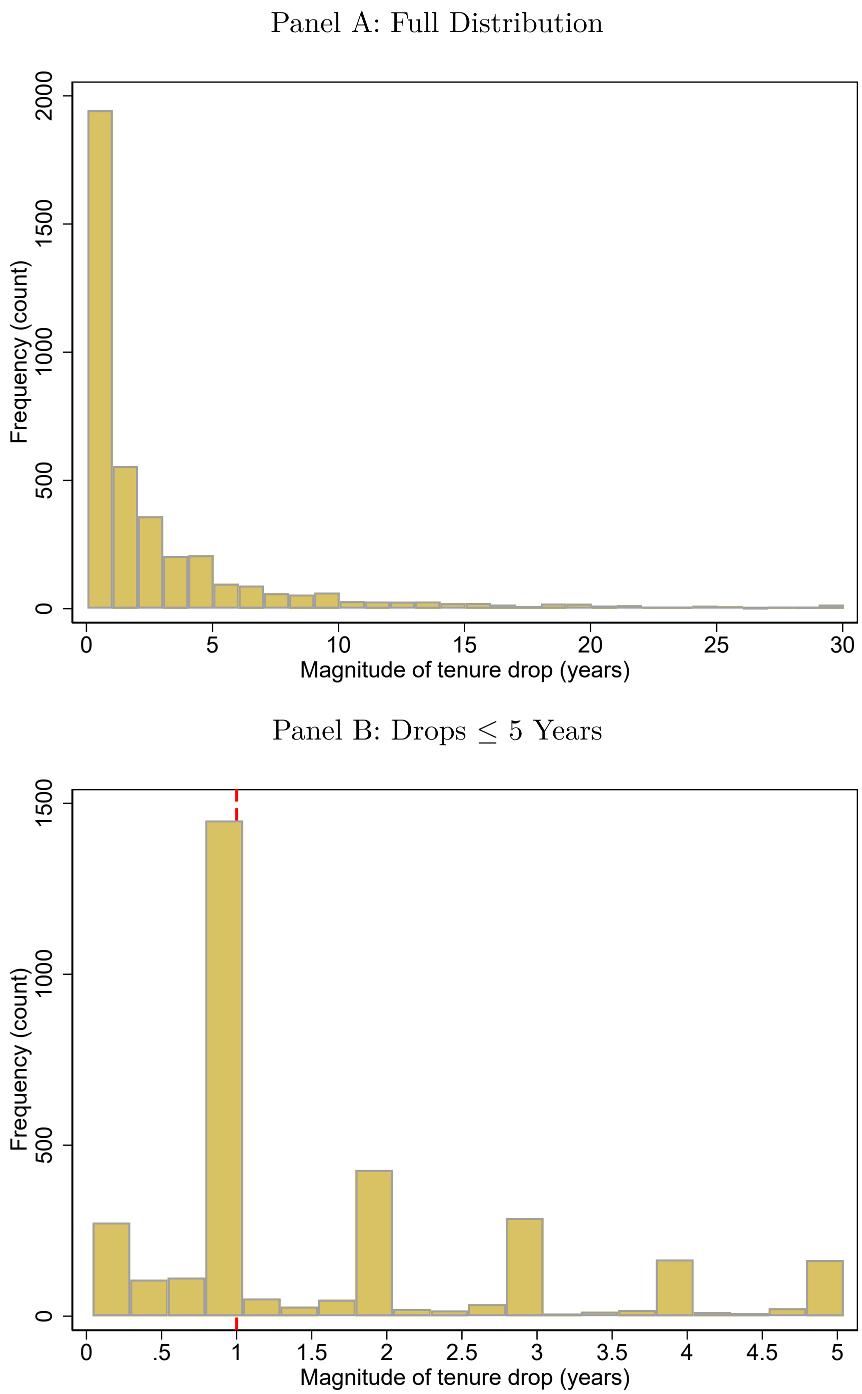
Notes.—Sources: Gallup Panel, 2023–2026. The table reports the coefficients associated with regressions of an indicator for whether the respondent uses AI frequently (daily or a few times a week; zero otherwise) on a vector of workplace perceptions, an indicator for whether the firm has communicated a clear AI strategy, an indicator for believing that AI is "very likely" or "somewhat likely" to replace their job in the next five years, and individual covariates and wave fixed effects. Workplace (management) practices are based on responses to the degree of agreement with the following statements on a 1–5 scale: (a) I feel connected to my organization's culture, (b) My organization consistently delivers on the promises it makes to customers, (c) I trust the leadership of this organization, (d) My organization cares about my overall wellbeing, (e) I have received meaningful feedback in the past week, and (f) I am treated with respect at work. Clear AI strategy is a binary in response to: Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." The wording for the job displacement question is: How likely is it that the job you have now will be eliminated within the next five years as a result of new technology, automation, robots or artificial intelligence?" Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Table A.6: Robustness: Workplace Determinants of AI Utilization (Full Distribution)

	Dep. var. = Use AI Often or Sometimes (1/0)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
C: Well-being	.273***							-.061
	[.024]							[.063]
C: Connected		.310***						.129**
		[.026]						[.054]
C: Feedback			.391***					.091*
			[.027]					[.048]
C: Customers				.117***				-.174***
				[.026]				[.047]
C: Trust					.221***			.170***
					[.030]			[.061]
C: Respect						.277***		.006
						[.027]		[.053]
Has Clear AI Strategy							3.082***	2.948***
							[.057]	[.091]
Age	-.023***	-.027***	-.023***	-.026***	-.032***	-.027***	-.018***	-.027***
	[.002]	[.002]	[.002]	[.002]	[.003]	[.002]	[.002]	[.003]
Some College	.393***	.319***	.470***	.310***	.184*	.439***	.349***	.127
	[.078]	[.082]	[.086]	[.082]	[.094]	[.084]	[.070]	[.109]
College or More	1.888***	1.732***	2.028***	1.760***	1.420***	1.983***	1.668***	1.341***
	[.073]	[.076]	[.080]	[.077]	[.089]	[.078]	[.067]	[.102]
Male	.289***	.297***	.270***	.283***	.323***	.269***	.272***	.223***
	[.053]	[.054]	[.059]	[.056]	[.062]	[.058]	[.049]	[.072]
White	-.403***	-.402***	-.375***	-.413***	-.384***	-.403***	-.269***	-.284***
	[.069]	[.071]	[.074]	[.072]	[.085]	[.073]	[.062]	[.090]
Black	.162	.095	.238**	.155	.161	.195*	.117	.261*
	[.104]	[.108]	[.115]	[.112]	[.130]	[.112]	[.092]	[.146]
Full time	.863***	.828***	.819***	.857***	.813***	.931***	.634***	.644***
	[.072]	[.077]	[.083]	[.078]	[.087]	[.078]	[.065]	[.109]
Has Children	.355***	.352***	.391***	.350***	.310***	.382***	.306***	.277***
	[.057]	[.059]	[.064]	[.060]	[.068]	[.063]	[.053]	[.080]
Is Married	.285***	.289***	.359***	.323***	.238***	.340***	.293***	.350***
	[.056]	[.058]	[.064]	[.060]	[.066]	[.063]	[.053]	[.079]
R-squared	.11	.11	.12	.11	.10	.12	.23	.22
Sample Size	71116	52733	43666	50814	34020	46478	69181	19699
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The table reports the coefficients associated with regressions of a 0–10 index of AI utilization on a vector of workplace perceptions, an indicator for whether the firm has communicated a clear AI strategy, individual covariates, and time fixed effects. The frequency index is constructed as follows: Daily=10, Few Times Week=8, Few Times Month=6, Few Times Year=4, Once a Year=2, Less than Once a Year=1, Never/Other=0. Workplace (management) practices are based on responses to the degree of agreement with the following statements on a 1–5 scale: (a) I feel connected to my organization's culture, (b) My organization consistently delivers on the promises it makes to customers, (c) I trust the leadership of this organization, (d) My organization cares about my overall wellbeing, (e) I have received meaningful feedback in the past week, and (f) I am treated with respect at work. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Figure A.4: Distribution of Tenure Drops Across Adjacent Survey Waves



Notes.—Sources: Gallup Panel, 2023–Q1:2026. The figure plots the distribution of negative wave-to-wave tenure changes among respondents observed in at least two consecutive waves. Tenure change is defined as $\Delta\text{Tenure}_{it} = \text{Tenure}_{it} - \text{Tenure}_{i,t-1}$; only observations with $\Delta\text{Tenure}_{it} < 0$ are included. Panel A displays the full distribution of the magnitude of tenure drops (i.e., $|\Delta\text{Tenure}_{it}|$) in one-year bins. Panel B restricts to drops of five years or fewer and uses quarter-year bins; the dashed red line marks the proposed cutoff of one year, below which tenure reductions are treated as measurement noise rather than employer transitions. The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

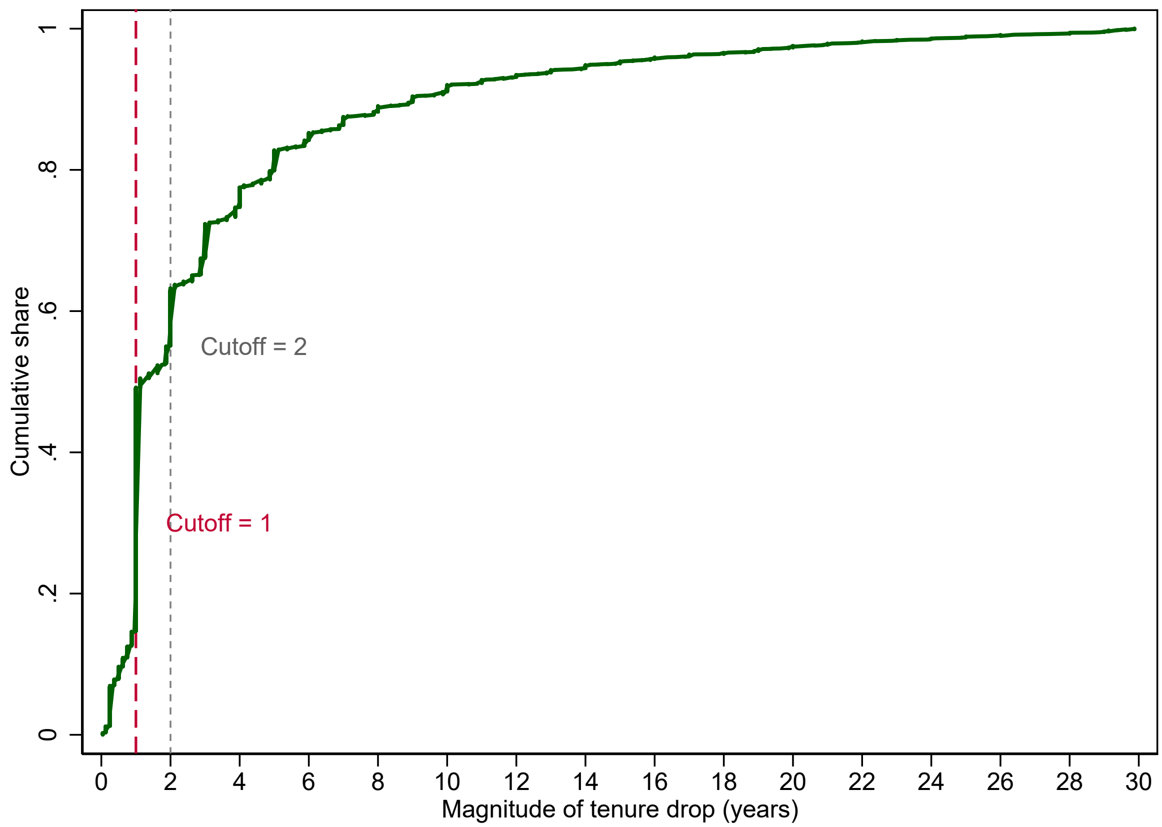
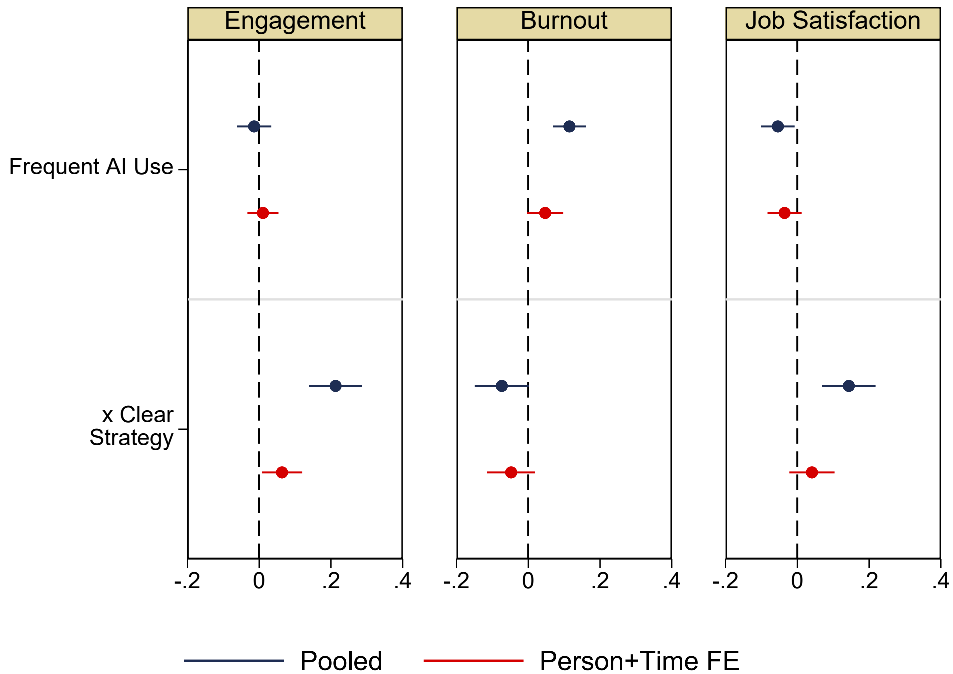


Figure A.5: Cumulative Distribution of Tenure Drops Across Adjacent Survey Waves

Notes.—Sources: Gallup Panel, 2023–Q1:2026. The figure plots the empirical cumulative distribution function of the magnitude of negative wave-to-wave tenure changes (i.e., $|\Delta\text{Tenure}_{it}|$ conditional on $\Delta\text{Tenure}_{it} < 0$). The dashed cranberry line marks a cutoff of one year; the short-dashed gray line marks a cutoff of two years. Tenure drops below the one-year cutoff are treated as measurement noise (e.g., rounding or within-employer recoding) rather than employer transitions; drops at or above the cutoff are classified as job switches. The two-year cutoff is shown as a robustness alternative. The sample is restricted to full- and part-time workers aged 20–65 observed in at least two consecutive waves, and the observations are weighted.

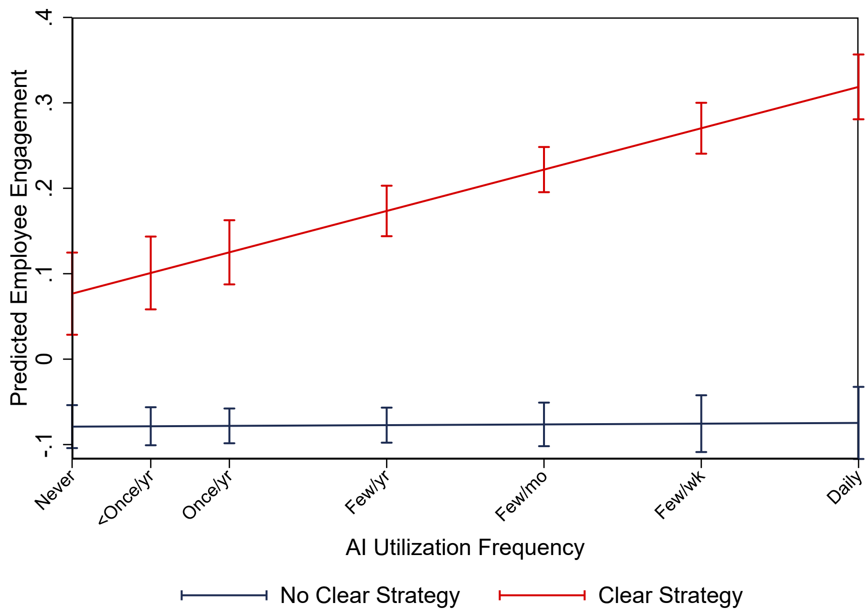
Figure A.6: Visualization of the Results on AI Complementarities with Clear AI Strategy



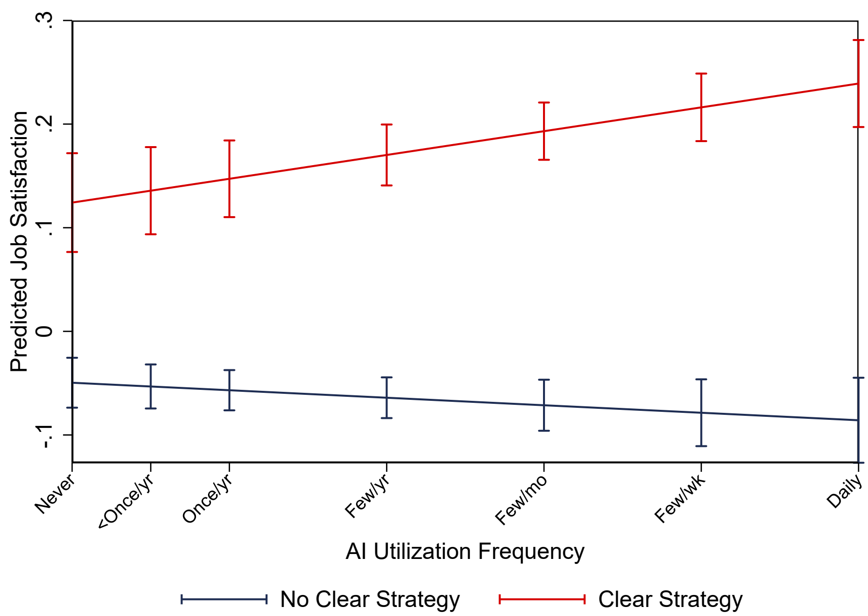
Notes.—Sources: Gallup Panel, 2023–Q1:2026. The figure plots coefficients on frequent AI use and its interaction with clear AI strategy from regressions of z-scores for employee engagement (Q4), burnout, and job satisfaction. Pooled estimates condition on demographics and no fixed effects; Person+Time FE estimates absorb person and time fixed effects. AI use – frequent is based on using AI daily or a few times a week; the normalization is relative to those who use it once a year, less than once a year, or never. Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices? (e.g., to increase productivity, efficiency, and quality)." Capped spikes denote 95% confidence intervals. Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.

Figure A.7: Predicted Employee Outcomes by AI Utilization Frequency and Clear AI Strategy

Panel A: Employee Engagement



Panel B: Job Satisfaction



Notes.—Sources: Gallup Panel, 2023–Q1:2026. The figure plots predicted values from regressions of standardized employee engagement (Panel A) and standardized job satisfaction (Panel B) on a continuous measure of AI utilization frequency interacted with an indicator for whether the organization has communicated a clear AI strategy, conditional on demographic controls and wave fixed effects. The AI utilization frequency index is constructed as: Daily = 10, Few Times/Week = 8, Few Times/Month = 6, Few Times/Year = 4, Once/Year = 2, Less than Once/Year = 1, Never = 0. Employee engagement is defined as a z-score of the average across four of the Q12 indicators due to missing values in the broader set: (Q1) I know what is expected of me at work, (Q3) At work, I have the opportunity to do what I do best every day, (Q6) There is someone at work who encourages my development, (Q7) At work, my opinions seem to count. Job satisfaction is a z-score of: "How satisfied are you with your place of employment as a place to work?" Clear AI strategy is a binary in response to: "Has your organization communicated a clear plan or strategy for integrating artificial intelligence (AI) technology or tools into current business practices?" Demographics include age, race (White, Black), full-time status, having children, being married, education (some college, college or more), and remote work status (always remote, sometimes remote). Standard errors are clustered at the individual level. The sample is restricted to full- and part-time workers aged 20–65, and the observations are weighted.